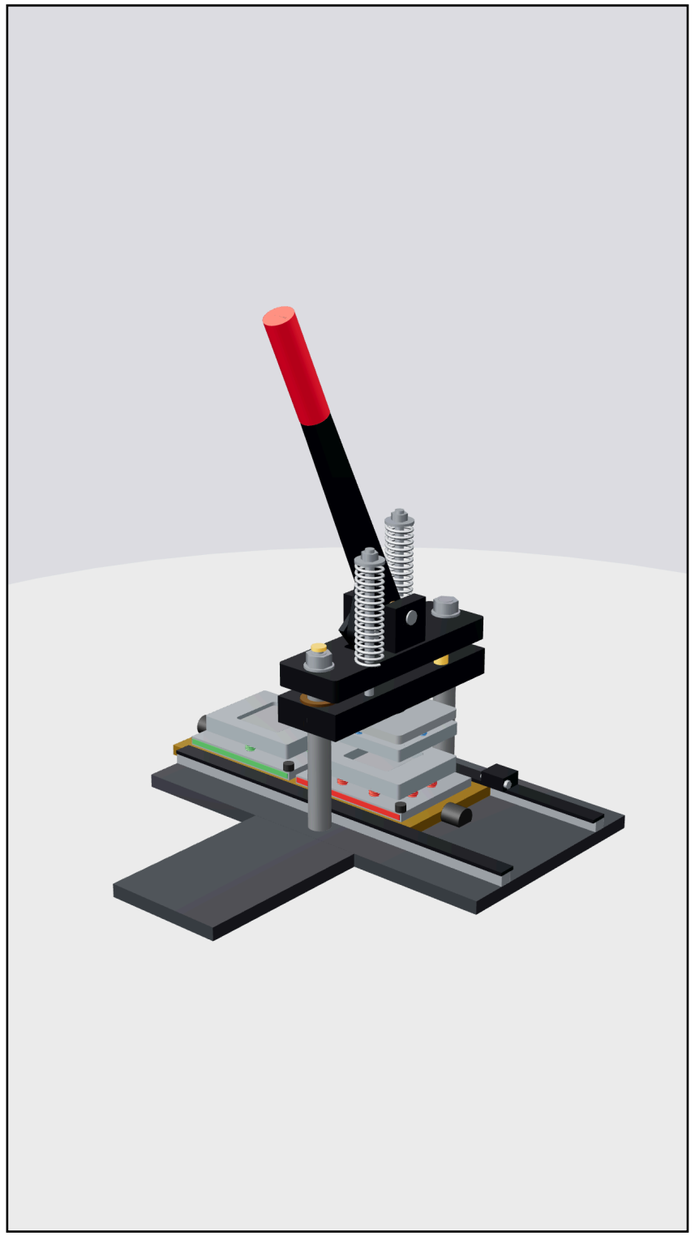
MagPress M4

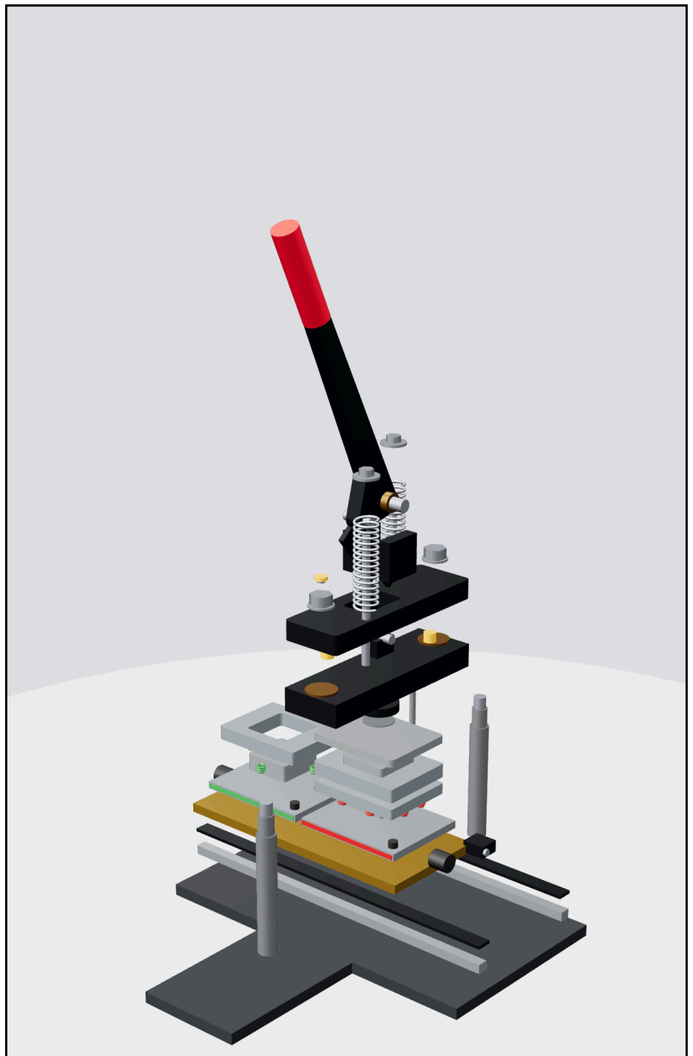
Manual photo fridge-magnet press · manufacturing drawing set · Rev B

4 swappable die sets: 50 × 50, 63.5 × 63.5, 53 × 80, 65 × 90 mm

DRAWING LIST

DRG NO.	TITLE	QTY	SHEET
MP-001	GENERAL ARRANGEMENT & HEIGHTS	—	2
MP-101	BASE PLATE	1	3
MP-102	GUIDE RAIL	2 (1 × -F, 1 × -B)	4
MP-103	KEEPER STRIP (GIB)	2 (1 × -F, 1 × -B)	5
MP-104	STATION STOP BLOCK	2	6
MP-105	DIE TABLE (SLIDE)	1	7
MP-106	DIE TABLE STOP PIN	1	8
MP-107	COLUMN	2	9
MP-108	CROWN	1	10
MP-109	RAM	1	11
MP-110	CLEVIS	1	12
MP-111	DIE HOLDER	1	13
MP-112	PIVOT CHEEK	2	14
MP-113	HANDLE LEVER (ONE PIECE)	1	15
MP-114	TOGGLE LINK	2	16
MP-116	PIN A / PIN B (TOGGLE)	2	17
MP-117	PIN O (HANDLE PIVOT)	1	18
MP-118	RETURN ROD	2	19
MP-119	PIVOT SPACER	2	20
MP-120	STATION PILOT	1	21
MP-201	UPPER DIE BODY (UNIVERSAL)	4 (1 per set)	22
MP-202	UPPER DIE SHANK	1 per die set (4)	23
MP-203	UPPER DIE PLUG	1 per die set (4)	24
MP-204	UPPER DIE FRAME	1 per die set (4)	25
MP-205	LOWER DIE SHOE A / B (UNIVERSAL)	A + B per set	26
MP-206	PICKUP PEDESTAL (DIE A)	1 per die set (4)	27
MP-207	PICKUP FRAME (DIE A)	1 per die set (4)	28
MP-208	CRIMP PEDESTAL (DIE B)	1 per die set (4)	29
MP-209	CRIMP FRAME (DIE B)	1 per die set (4)	30





Reference renderings (Rev B): assembled, handle up (left); exploded (right). Not for measurement.

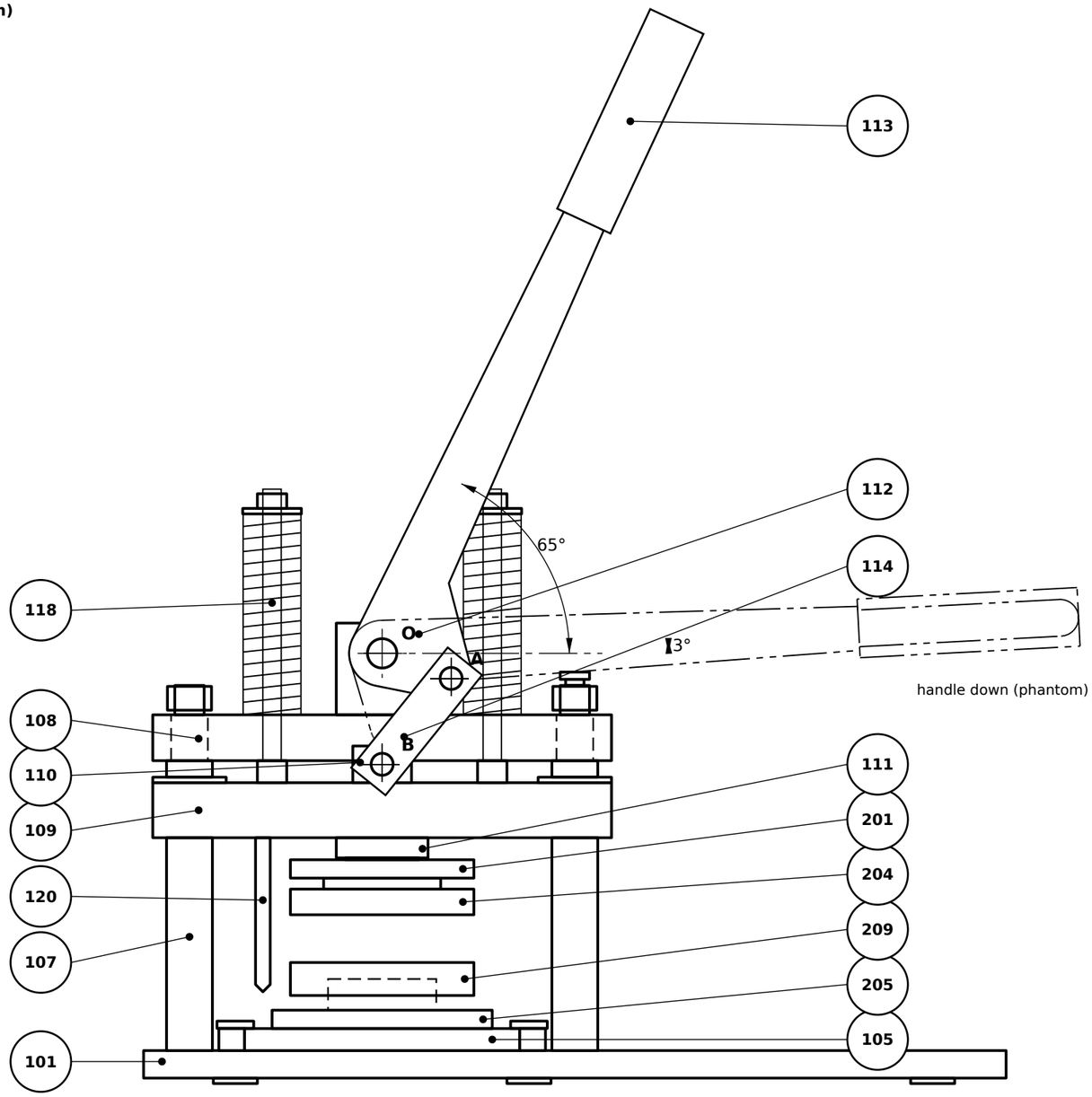
NOTES

1. Units mm. First-angle projection (ISO 128-30); every view is also labelled by name.
2. General tolerances ISO 2768-mK unless stated. Fits ISO 286. Threads ISO metric coarse 6H/6g unless stated. Ra 3.2 unless stated.
3. Column bores in base MP-101, crown MP-108 and ram MP-109 at 210.00 ±0.01: same datum scheme and CNC program.
4. Rev B die sets: bodies MP-201 and shoes MP-205 are common to all sizes. Plugs, upper frames, pedestals and lower frames are tabulated for 5050, 6363, 5380, 6590.
5. DXF profiles are supplied for cutting and wire-EDM. If a DXF and a drawing disagree, the drawing governs. Bought parts are in the build pack BOM.
6. Rev B (2026-10-27), failure-mode review: one-piece lever (MP-115 deleted), pin O Ø16 in PTFE-composite bushings, stronger return springs, station pilot MP-120 (new), universal die outers, A2 dies re-tempered after EDM, 15 mm base on pads, 16 kN design load.
7. Prove the die fits on a 7075 aluminium set before cutting tool steel.

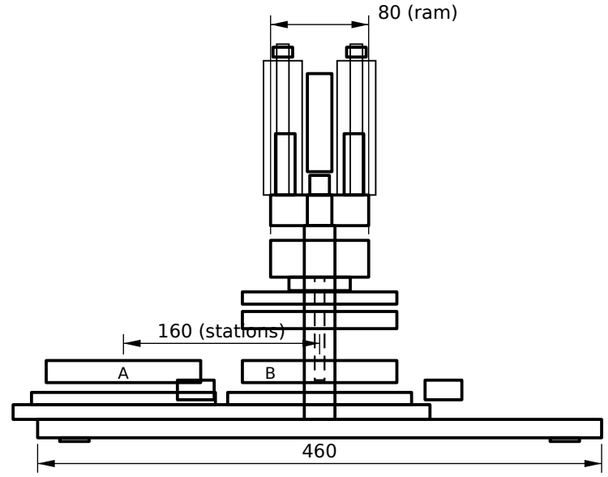
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 1 / 30
TITLE: **DRAWING SET — COVER & LIST**			
DRAWING NO.: **MP-000**	QTY: —	SCALE: —	UNITS: mm
MATERIAL: —			
HEAT TREATMENT / FINISH: —			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:

KEY HEIGHTS, handle up (mm from bench)

HEIGHT	LEVEL
586	grip top, handle up
324	return-rod top
234.5	pivot O
201	crown top
176	crown bottom
164	ram top
134	ram bottom
92	upper die bottom
66	lower die top
40	shoe top
30	die table top
18	base top
3	base underside (3 mm pads)
0	bench



SIDE ELEVATION (viewed from the left; front →)

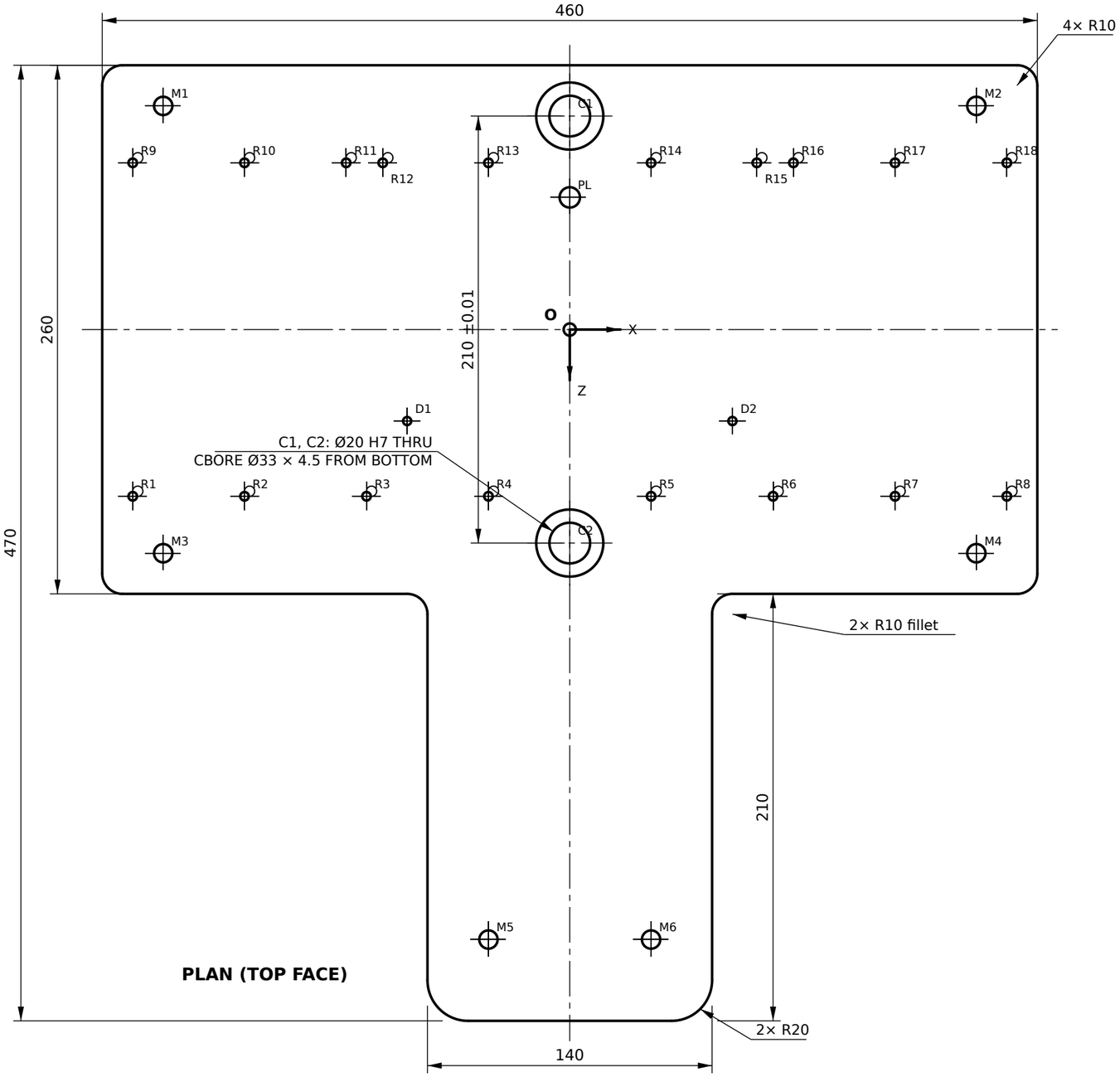


FRONT ELEVATION 1:5 (handle cut off)

NOTES

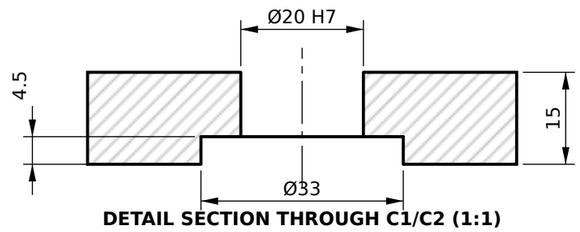
1. Shown with the handle UP (open), die table at "B under ram", die set 6363 (Rev B universal die outers 126 × 100). Heights from the bench top.
2. Toggle: crank O→A 40, link A→B 60, grip centre 320 from O (Rev A 290). Handle 65° → 3° above horizontal. Force gain 6.6 at the start, 34.6 at the end of the stroke.
3. Ram stroke 38.9. Dies meet after 26.0; working travel 12.9. Pickup stroke stops at 35.5. Station pilot MP-120 enters the table at 20.0, before the dies meet.
4. Open stop: 2 urethane bumpers on the ram top. Closed stop: adjustable M10 × 1 screw + urethane cap on the front column top, set so the lever rests at 3°.
5. Rev B: one-piece lever (MP-115 deleted); pin O Ø16; PTFE-composite bushings at O, A, B; return springs 60 N / 225 N; station pilot; 15 mm base on rubber pads.
6. Balloons = drawing numbers (MP-xxx). Pins MP-116 at A and B, MP-117 at O; spacers MP-119 on pin O; stop blocks MP-104 behind the table.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 2 / 30
TITLE: **GENERAL ARRANGEMENT & HEIGHTS**			
DRAWING NO.: **MP-001**	QTY: —	SCALE: 1:3 / 1:5	UNITS: mm
MATERIAL: —			
HEAT TREATMENT / FINISH: —			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED



HOLE TABLE

HOLE	X	Z	DESCRIPTION
C1	0	-105	Ø20 H7 THRU; CBORE Ø33 × 4.5 BOTTOM
C2	0	105	Ø20 H7 THRU; CBORE Ø33 × 4.5 BOTTOM
R1	-215	82	M5 THRU (rail screws)
R2	-160	82	M5 THRU (rail screws)
R3	-100	82	M5 THRU (rail screws)
R4	-40	82	M5 THRU (rail screws)
R5	40	82	M5 THRU (rail screws)
R6	100	82	M5 THRU (rail screws)
R7	160	82	M5 THRU (rail screws)
R8	215	82	M5 THRU (rail screws)
R9	-215	-82	M5 THRU (rail screws)
R10	-160	-82	M5 THRU (rail screws)
R11	-110	-82	M5 THRU (rail screws)
R12	-92	-82	M5 THRU (rail screws)
R13	-40	-82	M5 THRU (rail screws)
R14	40	-82	M5 THRU (rail screws)
R15	92	-82	M5 THRU (rail screws)
R16	110	-82	M5 THRU (rail screws)
R17	160	-82	M5 THRU (rail screws)
R18	215	-82	M5 THRU (rail screws)
M1	-200	-110	Ø9 THRU (bench mounting, M8)
M2	200	-110	Ø9 THRU (bench mounting, M8)
M3	-200	110	Ø9 THRU (bench mounting, M8)
M4	200	110	Ø9 THRU (bench mounting, M8)
M5	-40	300	Ø9 THRU (bench mounting, M8)
M6	40	300	Ø9 THRU (bench mounting, M8)
D1	-80	45	Ø4 × 90° spot, 1.5 deep (detent)
D2	80	45	Ø4 × 90° spot, 1.5 deep (detent)
PL	0	-65	Ø10 × 8 deep (pilot clearance)



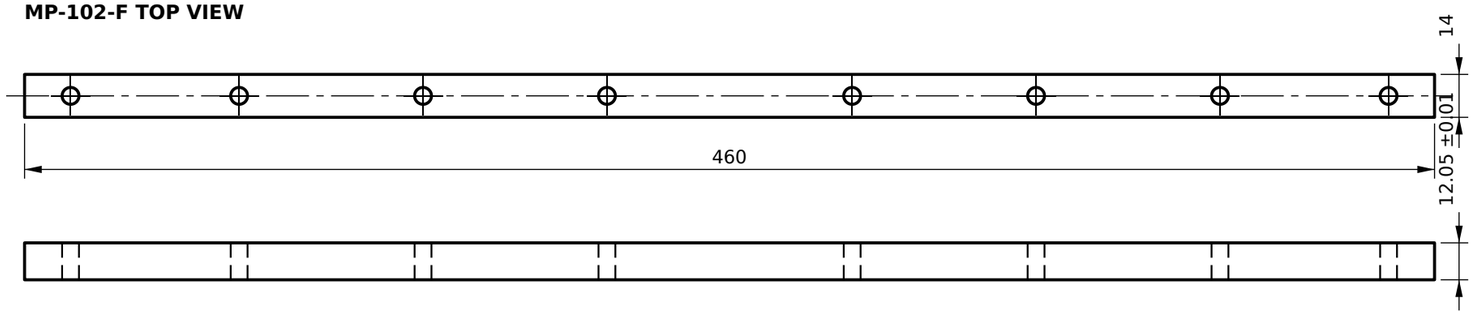
DETAIL SECTION THROUGH C1/C2 (1:1)

NOTES

1. Rev B: 15 mm S355 (was 12 S235). At the 16 kN design load: 112 MPa, 0.26 mm (Rev A: 175 MPa, 0.51 mm). 0.08 mm at a normal 5 kN stroke.
2. Face both sides to 15.0 ±0.1. Flatness 0.05 over the slide area X ±230 × Z ±75, Ra 1.6.
3. C1/C2 Ø20 H7 at 210.00 ±0.01: same datum and program as crown MP-108 and ram MP-109. CBORE Ø33 × 4.5 from the BOTTOM.
4. M1–M6: Ø9 THRU bench-mounting holes (M8). Underside: five 3 mm adhesive rubber pads Ø24 at (±200, ±80) and (0, 300).
5. PL: Ø10 × 8 deep clearance for the station pilot MP-120. D1/D2: Ø4 × 90° spots, 1.5 deep (table detents).
6. Hole positions from datum O (midpoint of C1–C2). X to the right; Z toward the front (tongue).
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 3 / 30
TITLE: **BASE PLATE**			
DRAWING NO.: **MP-101**	QTY: 1	SCALE: 1:2.5	UNITS: mm
MATERIAL: S355J2 steel plate 20 mm, profile cut, faced to 15.0			
HEAT TREATMENT / FINISH: Slide area: QPQ nitrocarburize (prototype: black oxide + oil). Rest: black powder coat, bores masked			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

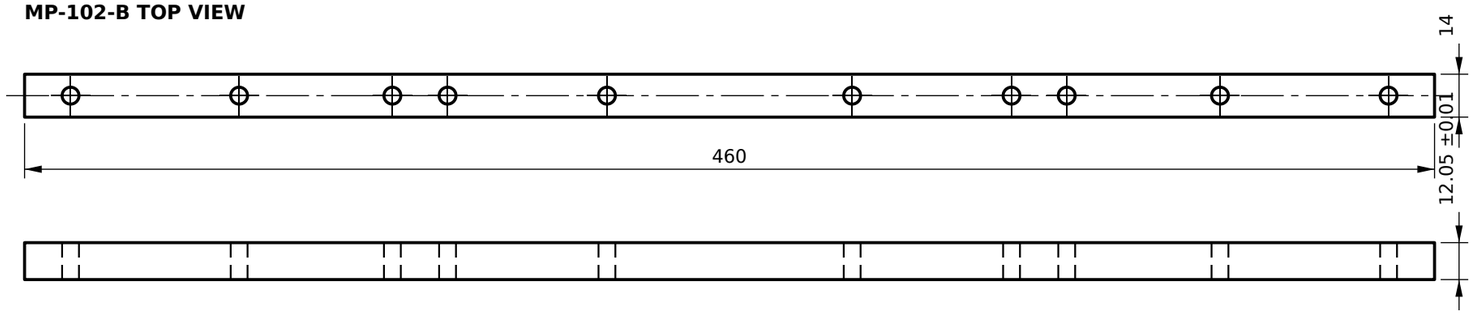
MP-102-F TOP VIEW



Hole X positions (from rail centre), all Ø5.5 THRU:

-215, -160, -100, -40, 40, 100, 160, 215

MP-102-B TOP VIEW



Hole X positions (from rail centre), all Ø5.5 THRU:

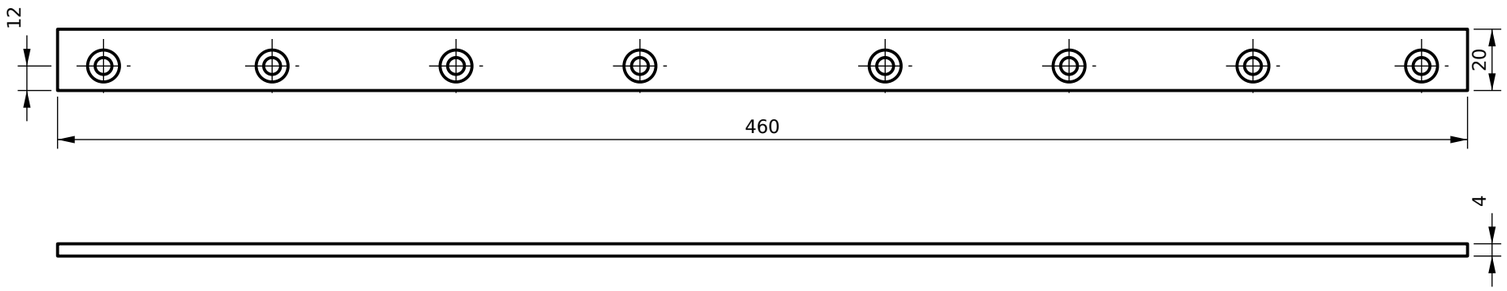
-215, -160, -110, -92, -40, 40, 92, 110, 160, 215

NOTES

1. Height 12.05 ±0.01: equal to die table MP-105 thickness + 0.05 running clearance. Grind both rails together.
2. Ground faces: top, bottom and inner face (faces the die table). Inner face straight within 0.02 over 460.
3. Variant -F: front rail, 8 holes. Variant -B: back rail, 10 holes (includes stop-block screws).
4. Holes Ø5.5 THRU on the rail centreline; hole X positions measured from the rail centre.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 4 / 30
TITLE: **GUIDE RAIL**			
DRAWING NO.: **MP-102**	QTY: 2 (1 × -F, 1 × -B)	SCALE: 1:2	UNITS: mm
MATERIAL: C45 bright flat 16 × 14, ground to 14 × 12.05			
HEAT TREATMENT / FINISH: Black oxide; ground faces oiled			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:

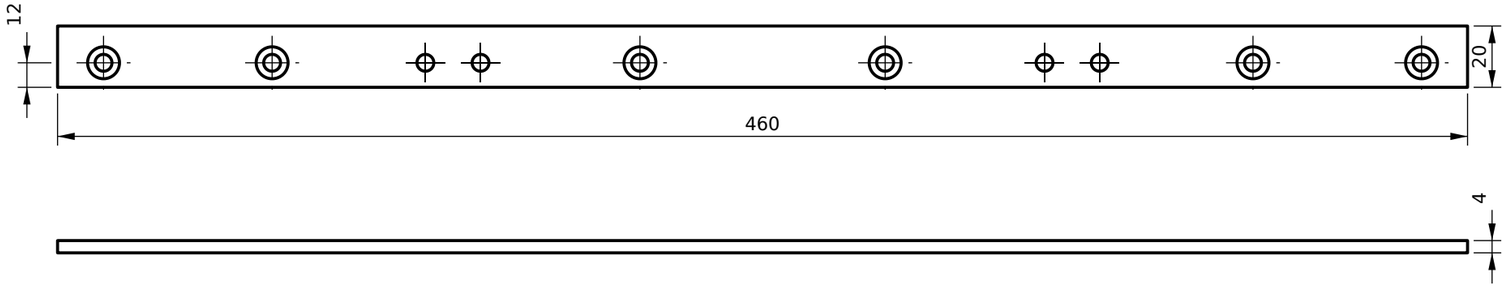
MP-103-F TOP VIEW (inner edge at bottom)



Countersunk (M5 ISO 10642):

-215, -160, -100, -40, 40, 100, 160, 215

MP-103-B TOP VIEW (inner edge at bottom)



Countersunk (M5 ISO 10642):

-215, -160, -40, 40, 160, 215

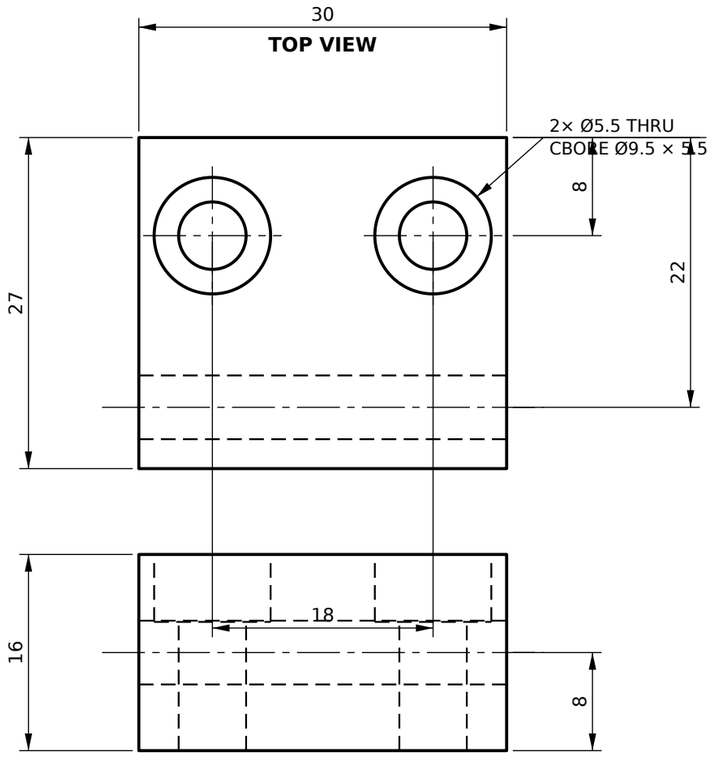
Plain Ø5.5 THRU:

-110, -92, 92, 110

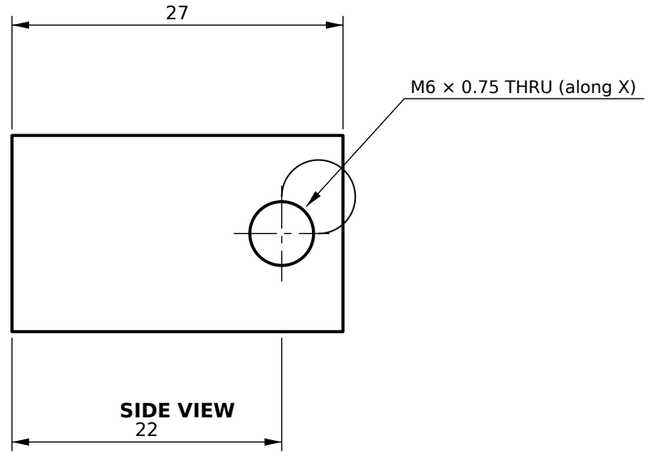
NOTES

1. Sits on top of the guide rail and overlaps the die table edge by 5 mm (inner edge).
2. Countersunk holes: Ø5.5 THRU, 90° countersink to Ø10.4 (for M5 ISO 10642), from top.
3. Variant -B: four plain Ø5.5 holes at X ±92 and ±110 for the stop-block screws (no countersink).
4. Hole line 12.0 from the inner edge. Hole X positions measured from the strip centre.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 5 / 30
TITLE: **KEEPER STRIP (GIB)**			
DRAWING NO.: **MP-103**	QTY: 2 (1 × -F, 1 × -B)	SCALE: 1:2	UNITS: mm
MATERIAL: C45 bright flat 20 × 4			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:



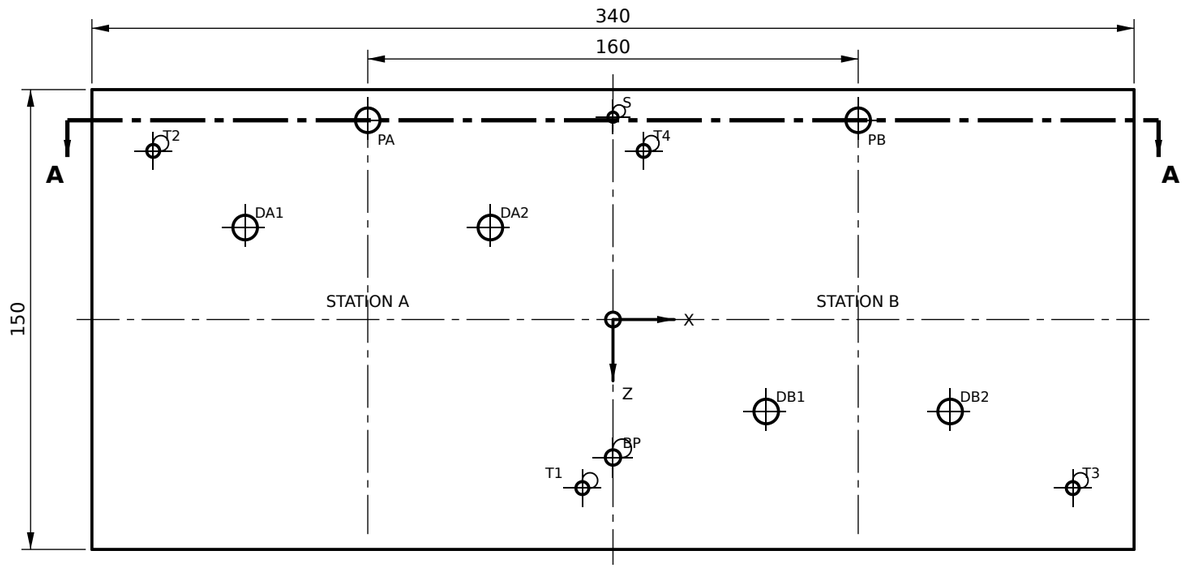
FRONT VIEW (from the die table side)



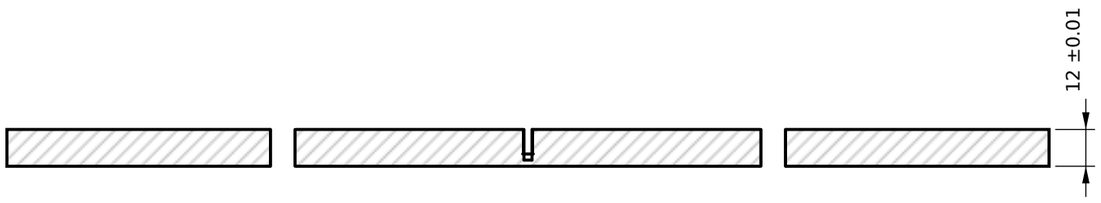
NOTES

1. Bolted through keeper MP-103-B and rail MP-102-B into the base with 2 × M5 × 35 ISO 4762 + threadlocker.
2. Stop screw: M6 × 0.75 × 25 hex screw + thin lock nut (bought). Coarse stop only: the pilot MP-120 sets the exact station position.
3. The 7 mm overhang (Z 20–27) sits above the die table; the table stop pin MP-106 hits the screw tip.
4. Z measured from the back face (the face over the keeper).
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

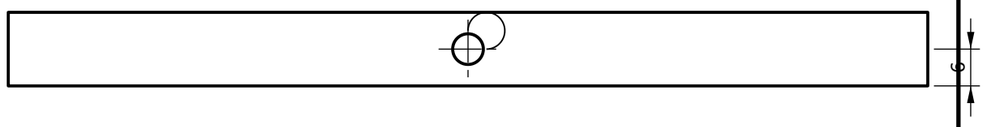
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 6 / 30
TITLE: **STATION STOP BLOCK**			
DRAWING NO.: **MP-104**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: C45 bright 30 × 16 flat			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED



PLAN



SECTION A–A (through the pilot holes)



END VIEW (K1/K2) 1:1

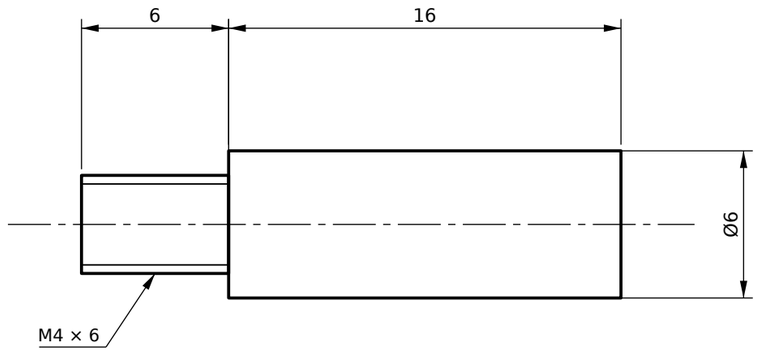
HOLE TABLE

HOLE	X	Z	DESCRIPTION
DA1	-120	-30	Ø8 H7 × 10 deep (shoe dowel)
DA2	-40	-30	Ø8 H7 × 10 deep (shoe dowel)
DB1	50	30	Ø8 H7 × 10 deep (shoe dowel)
DB2	110	30	Ø8 H7 × 10 deep (shoe dowel)
T1	-10	55	STI M5 THRU for Helicoil (thumb screw)
T2	-150	-55	STI M5 THRU for Helicoil (thumb screw)
T3	150	55	STI M5 THRU for Helicoil (thumb screw)
T4	10	-55	STI M5 THRU for Helicoil (thumb screw)
S	0	-66	M4 × 8 deep (stop pin MP-106)
BP	0	45	M6 THRU (ball plunger)
PA	-80	-65	Ø8 H8 THRU (station pilot), ±0.02
PB	80	-65	Ø8 H8 THRU (station pilot), ±0.02
K1	-170	0	M6 × 12 in end face, Y = 6 (knob)
K2	170	0	M6 × 12 in end face, Y = 6 (knob)

NOTES

1. Thickness 12.00 ±0.01 and flat within 0.03 after anodising. Guide rails MP-102 are 12.05.
2. Rev B: PA/PB Ø8 H8 THRU take the station pilot MP-120, ±0.02 to the shoe dowels. Rev B shoe pattern: dowels and thumb holes moved for the 150 × 120 universal shoes.
3. DA/DB: Ø8 H7 × 10 (shoe dowels; A and B use different pitches so the shoes cannot be swapped). T1–T4: STI M5 THRU for Helicoils.
4. S: M4 × 8 (stop pin MP-106). BP: M6 THRU (ball plunger). K1/K2: M6 × 12 in each end face, centred (push knobs).
5. Hole positions from the table centre. Station centres at X = −80 (A) and +80 (B), Z = 0.
6. Long edges: 0.5 × 45° chamfer only (they run against the rails).
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **B**	SHEET: 7 / 30	
TITLE: **DIE TABLE (SLIDE)**			
DRAWING NO.: **MP-105**	QTY: 1	SCALE: 1:2	UNITS: mm
MATERIAL: EN AW-6061-T6 plate 15 mm, faced to 12.00			
HEAT TREATMENT / FINISH: Hard anodise 25 µm, gold, PTFE-sealed; fit Helicoils after anodising			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

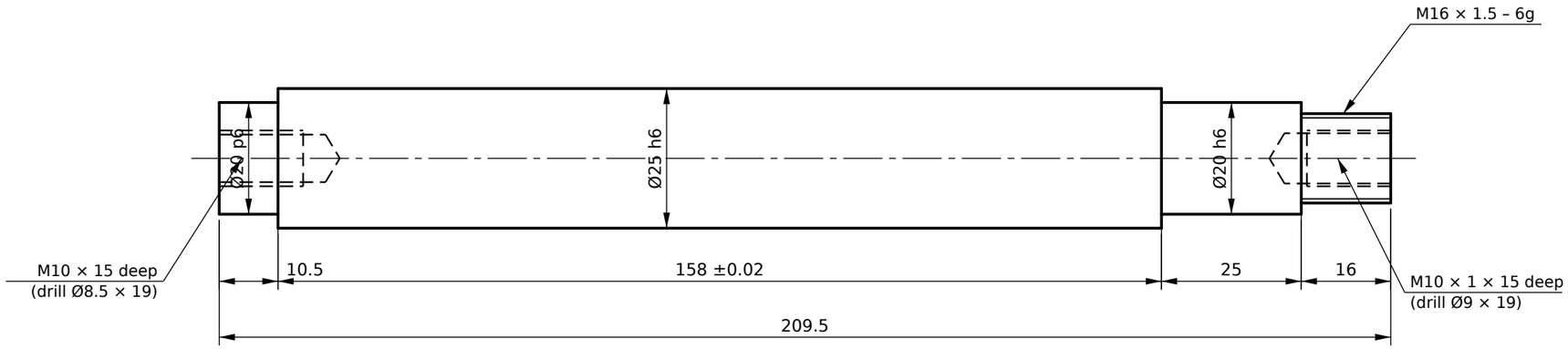


SIDE VIEW

NOTES

1. Screwed into the die table (MP-105, hole S) with Loctite 243.
2. Contact face Ø6 must be square to the axis within 0.05.
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 8 / 30
TITLE: **DIE TABLE STOP PIN**			
DRAWING NO.: **MP-106**	QTY: 1	SCALE: 4:1	UNITS: mm
MATERIAL: C45 Ø8 bright bar			
HEAT TREATMENT / FINISH: Harden 45–50 HRC; black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

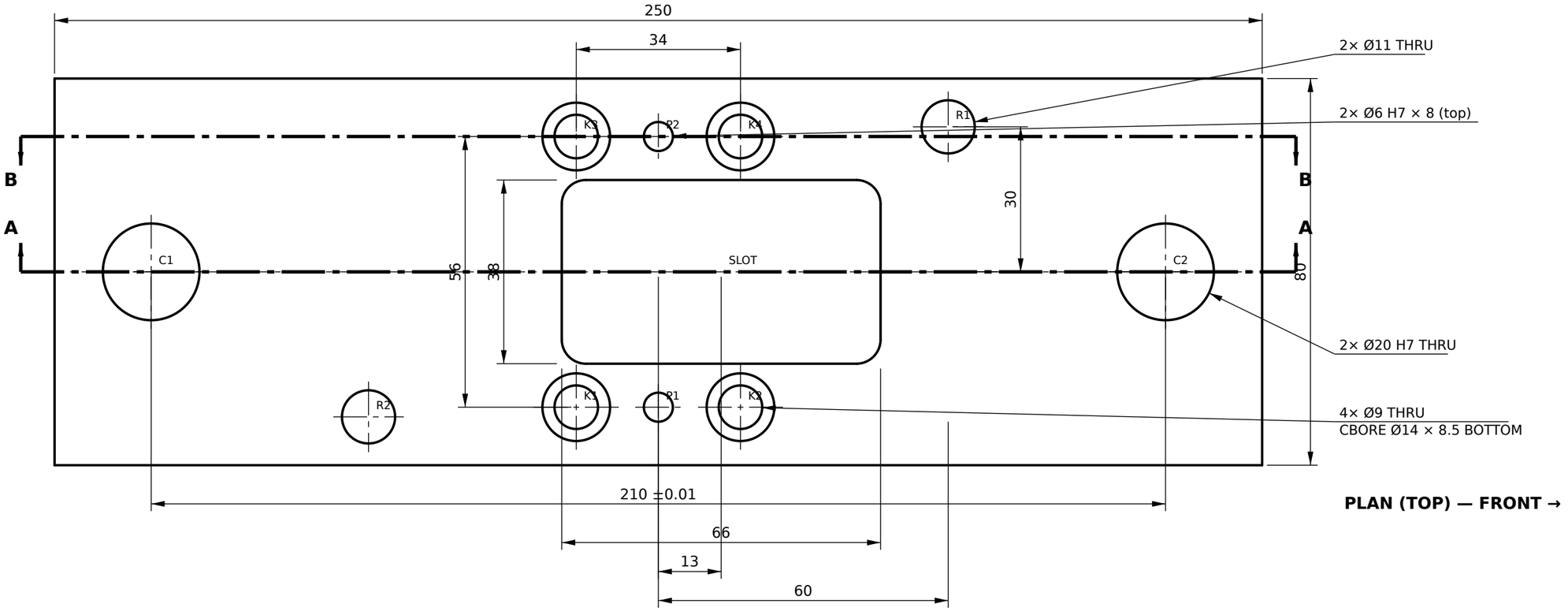


SIDE VIEW

NOTES

1. Ø25 h6 length 158.00 ±0.02; the two columns matched within 0.01 (sets crown parallelism).
2. Rev B: bottom spigot Ø20 p6 × 10.5 for the 15 mm base. Clamp from below: M10 × 25 ISO 4762 + washer Ø32 × 4, 40 N·m.
3. Ø20 h6 top spigot passes through the crown (Ø20 H7). Nut M16 × 1.5 + Nord-Lock washer on top, 100 N·m.
4. Rev B: top end M10 × 1 × 15 (both columns). Front column takes the adjustable handle stop: M10 × 1 set screw + jam nut + urethane cap.
5. Shoulders square to the axis within 0.01. Run-out of all diameters 0.01 to the Ø25 axis.
6. Chamfers: ends 1 × 45°; steps 0.5 × 45°. Thread undercut DIN 76-A at the M16 thread end.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

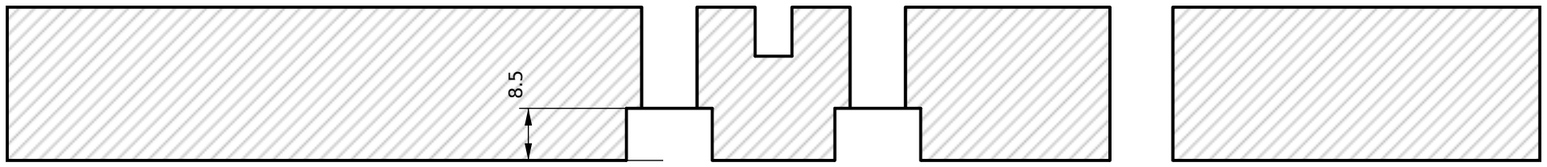
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 9 / 30
TITLE: **COLUMN**			
DRAWING NO.: **MP-107**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: 100Cr6 (52100 / GCr15) Ø30 bar			
HEAT TREATMENT / FINISH: Ø25 section: induction harden 58–62 HRC, hard chrome 8–12 µm, ground to h6, Ra 0.2			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED



PLAN (TOP) — FRONT →



SECTION A–A

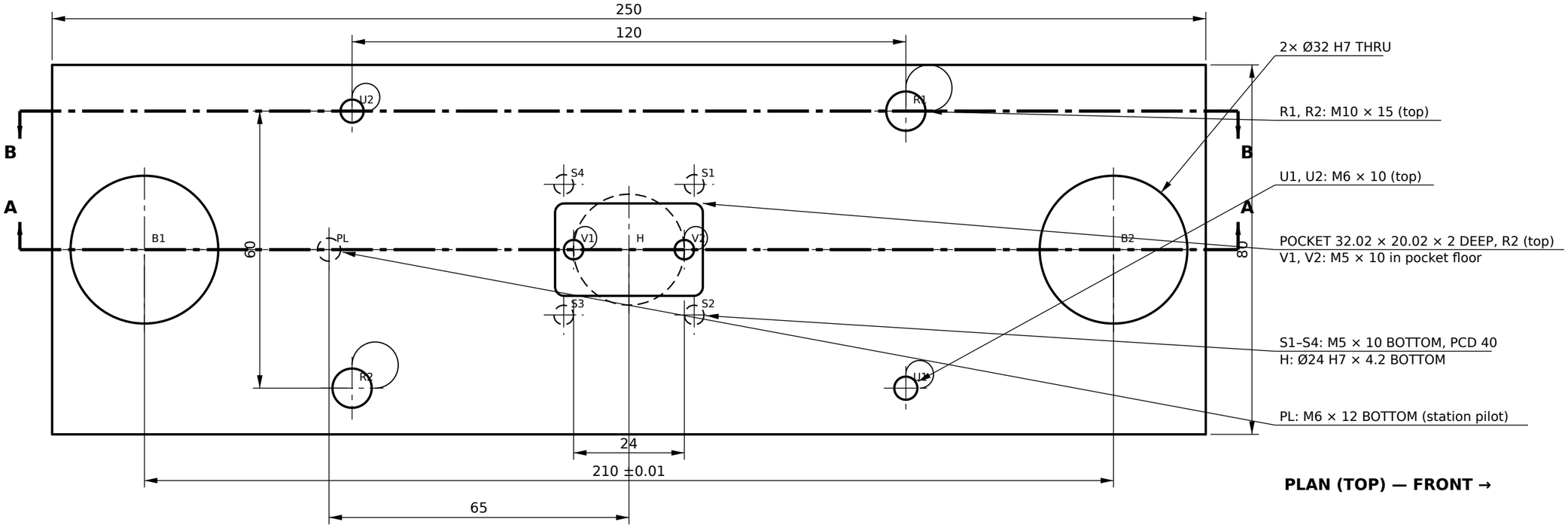


SECTION B–B (X = +28)

NOTES

1. C1/C2 Ø20 H7 at 210.00 ±0.01: same datum and program as base MP-101 and ram MP-109.
2. Rev B: return-rod holes R1/R2 moved to X = ±30. Design load raised to 16 kN: 192 MPa (C45 N, Re ≈ 340), 0.27 mm.
3. K1–K4: Ø9 THRU, CBORE Ø14 × 8.5 from the BOTTOM (M8 × 30 ISO 4762 + Nord-Lock into the cheeks MP-112).
4. P1/P2: Ø6 H7 × 8 from the top (cheek dowels). Slot 38 × 66, R5, THRU (links pass through).
5. Top and bottom faces parallel 0.02, flat 0.02. Plan drawn with the front of the press to the right.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 10 / 30
TITLE: **CROWN**			
DRAWING NO.: **MP-108**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: C45 flat 80 × 30, faced to 25			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:



PLAN (TOP) — FRONT →

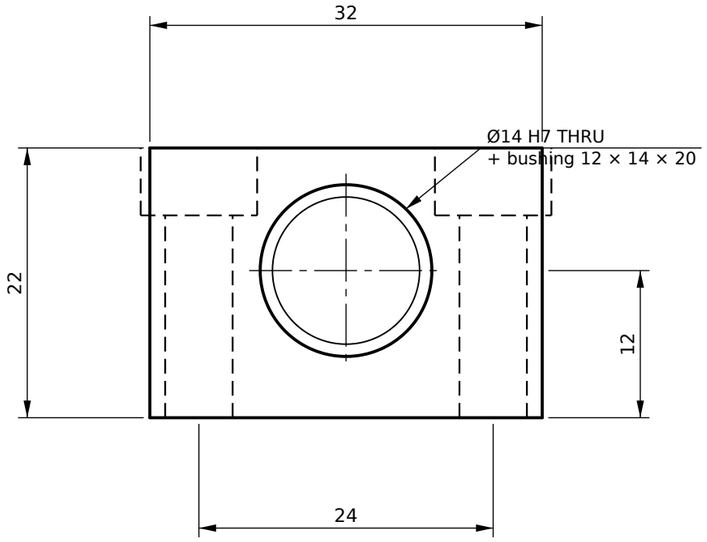
SECTION A–A

SECTION B–B (X = +30)

NOTES

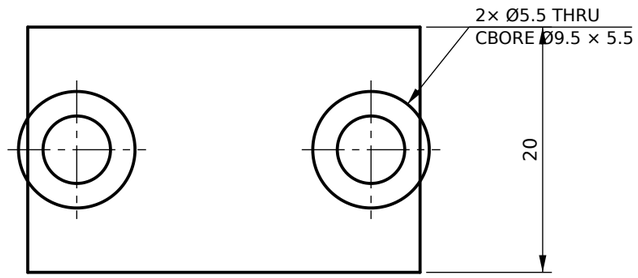
1. Rev B: width 80 (was 70) to carry the return rods at X = ±30 with full thread walls.
2. B1/B2 Ø32 H7 at 210.00 ±0.01 for flanged self-lubricating bronze bushings ID 25 / OD 32 / L 33; same program as MP-101/MP-108.
3. Top: clevis pocket 20.02 × 32.02 × 2.0 (R2) with V1/V2 M5 × 10; R1/R2 M10 × 15 (return rods); U1/U2 M6 × 10 (urethane bumpers).
4. Bottom: H Ø24 H7 × 4.2 (die-holder spigot); S1–S4 M5 × 10 on PCD 40 at 45°; Rev B: PL M6 × 12 for the station pilot MP-120.
5. Top and bottom faces parallel 0.02. Plan drawn with the front of the press to the right; hidden features are on the bottom.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 11 / 30
TITLE: **RAM**			
DRAWING NO.: **MP-109**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: C45 flat 90 × 35, faced to 80 × 30			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:



FRONT VIEW (pin axis = X)

TOP VIEW

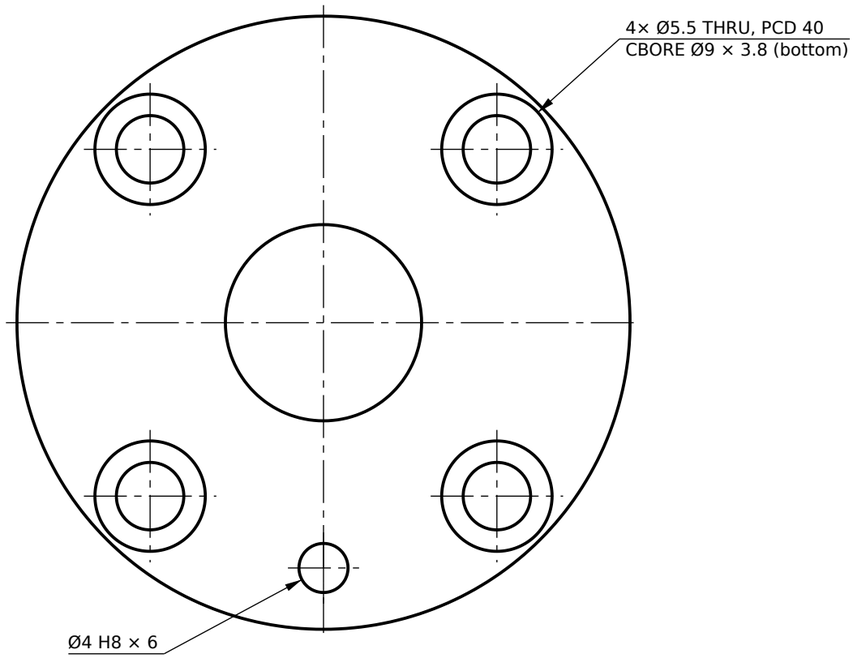


NOTES

1. Sits 2 mm deep in the ram pocket; 2 × M5 × 25 ISO 4762 + threadlocker.
2. Rev B: bore Ø14 H7 with a PTFE-composite plain bushing (DU / PAP type) 12 × 14 × 20 pressed in; pin B Ø12 runs dry.
3. Bore axis parallel to the bottom face within 0.02 and square to the long sides within 0.02.
4. Vertical edges chamfer 2.5 × 45° to clear the R2 pocket corners.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 12 / 30
TITLE: **CLEVIS**			
DRAWING NO.: **MP-110**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: C45 flat 25 × 35			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

SECTION THROUGH X AXIS

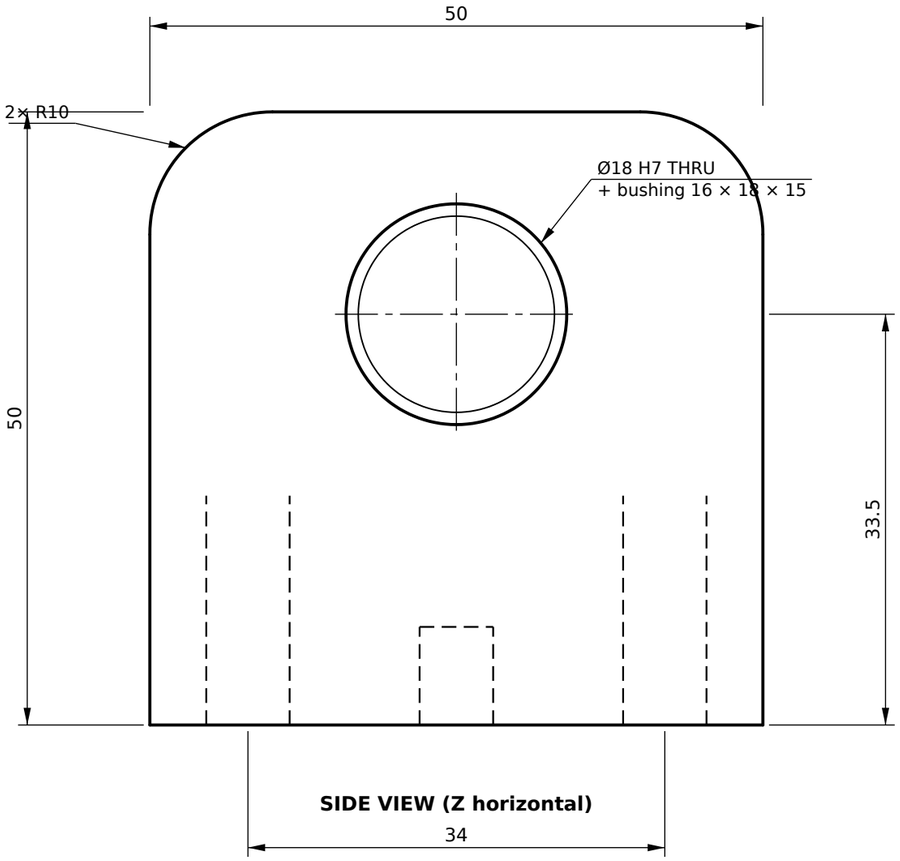


BOTTOM VIEW (front at bottom)

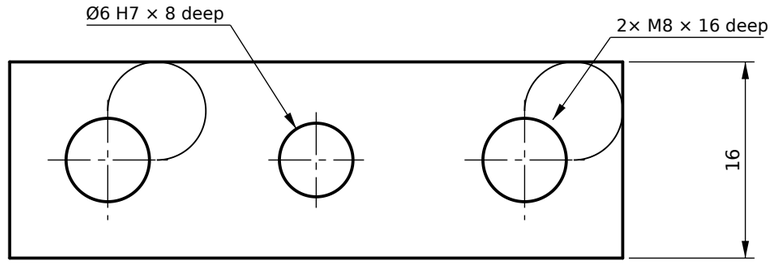
NOTES

1. Spigot Ø24 h6 locates in the ram (Ø24 H7). 4 × M5 × 16 DIN 7984 low-head screws from below + threadlocker.
2. Bore Ø16 H7 takes the upper-die shank MP-202. Bottom face (die seat) square to the bore within 0.01, flat 0.01.
3. Lock screw: M5 × 10 ISO 4027 cone-point set screw with nylon patch; seats in the shank V-groove. Check at every die change.
4. Orientation hole Ø4 H8 × 6 deep in the bottom face at 20 mm radius, on the Z axis toward the FRONT.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 13 / 30
TITLE: **DIE HOLDER**			
DRAWING NO.: **MP-111**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: 42CrMo4 Ø55 bar, QT 28–32 HRC			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN: / CHECKED:



SIDE VIEW (Z horizontal)



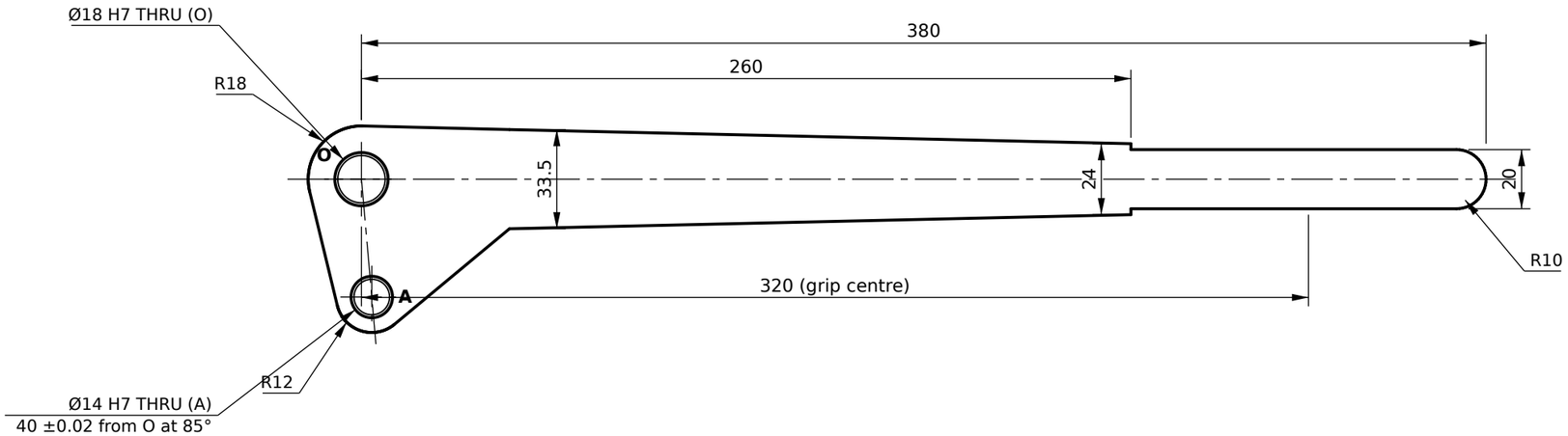
BOTTOM VIEW

NOTES

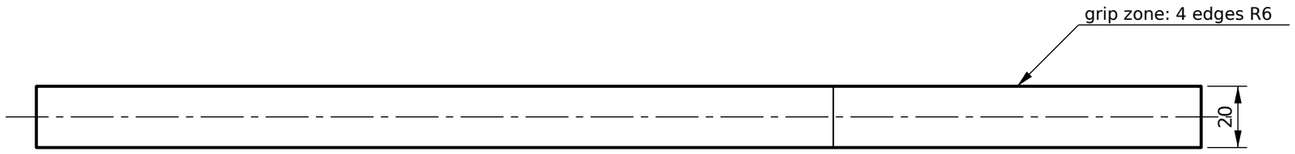
1. Rev B: pivot bore Ø18 H7 for a PTFE-composite plain bushing (DU / PAP type) 16 × 18 × 15 (pin O is now Ø16). Line-ream both cheeks together after fitting to the crown.
2. Fixed to the crown with 2 × M8 × 30 ISO 4762 + Nord-Lock washers from below + 1 × Ø6 × 16 dowel. Carries the full toggle reaction (16 kN design).
3. Bottom face flat 0.02, square to the sides.
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 14 / 30
TITLE: **PIVOT CHEEK**			
DRAWING NO.: **MP-112**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: C45 flat 20 × 55			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PROFILE (handle axis horizontal)



EDGE VIEW (thickness)

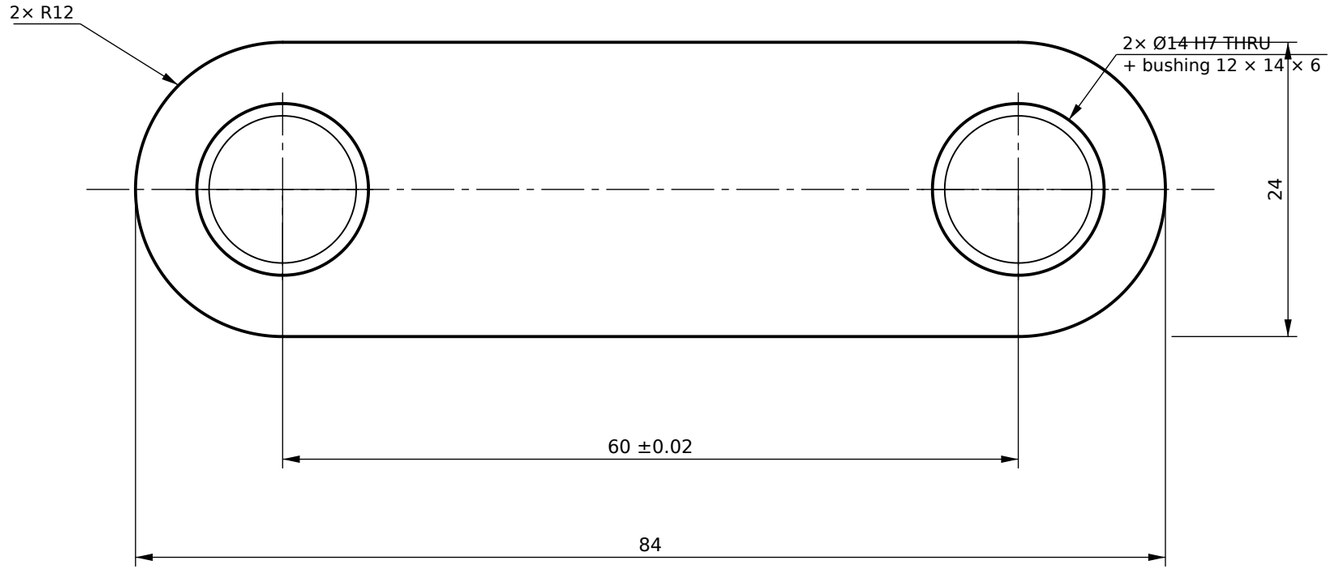


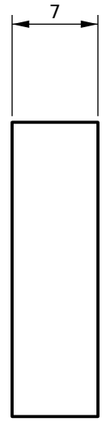
NOTES

1. Rev B: replaces the Rev A hub + screwed bar (MP-115 deleted). The Rev A M12 thread reached ≈ 465 MPa at a 200 N push; this section sees ≈ 14 MPa (36 MPa at 500 N).
2. Profile from DXF (laser/waterjet + CNC, or wire EDM). Faces parallel 0.05.
3. O: Ø18 H7 for a PTFE-composite plain bushing (DU / PAP type) 16 × 18 × 20. A: Ø14 H7 for a 12 × 14 × 20. Crank O→A = 40.00 ±0.02 at 85° ±0.1° to the handle axis.
4. Grip zone (u 260–380): 20 × 20 with all four long edges R6; red TPR grip Ø32 / Ø24 × 120 pushed on with rubber adhesive. Grip centre 320 from O.
5. Handle stop: the lever underside rests on the adjustable stop screw on the front column at 3° above horizontal.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 15 / 30
TITLE: **HANDLE LEVER (ONE PIECE)**			
DRAWING NO.: **MP-113**	QTY: 1	SCALE: 1:2	UNITS: mm
MATERIAL: C45 plate 25 mm, faced to 20			
HEAT TREATMENT / FINISH: Black oxide; grip zone edges R6			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PROFILE



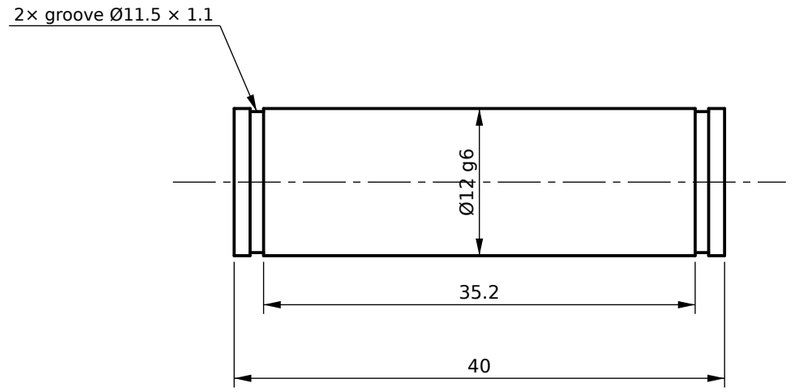


EDGE VIEW

NOTES

1. Hole centres 60.00 ±0.02; make both links as a matched pair (within 0.01).
2. Rev B: bores Ø14 H7 with PTFE-composite plain bushing (DU / PAP type) 12 × 14 × 6 pressed flush. Pins run dry.
3. Faces parallel 0.02. Stamp the pair with matching numbers.
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 16 / 30
TITLE: **TOGGLE LINK**			
DRAWING NO.: **MP-114**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: C45 plate 8 mm, faced to 7			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

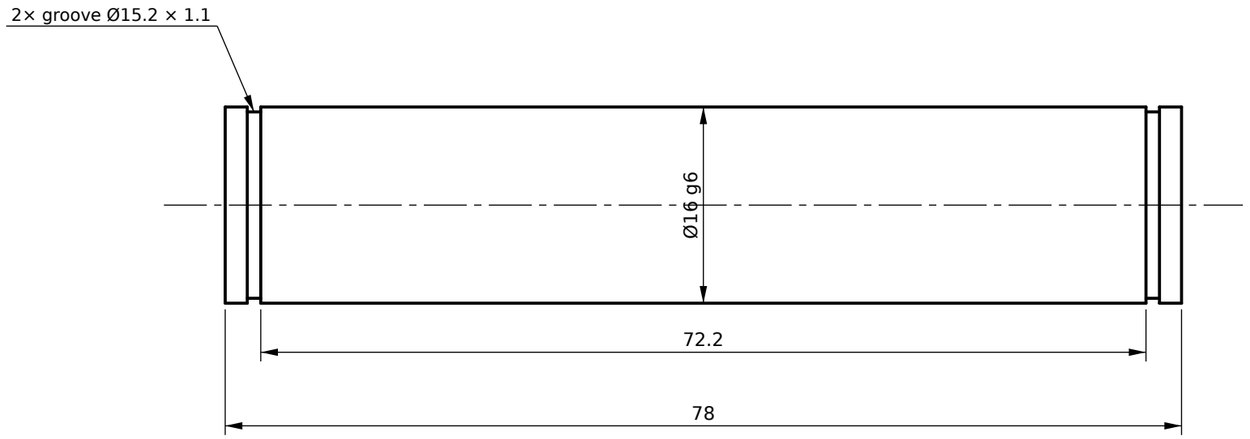


SIDE VIEW

NOTES

1. Retaining-ring grooves for DIN 471 – 12: groove Ø11.5 h12, width 1.1 H13, both ends.
2. Ends 0.5 × 45° chamfer. Groove inner edge at 17.6 from the pin centre (bearing faces at ±17.5).
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 17 / 30
TITLE: **PIN A / PIN B (TOGGLE)**			
DRAWING NO.: **MP-116**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: 100Cr6 (GCr15) bar, or hardened ground dowel stock Ø12 g6			
HEAT TREATMENT / FINISH: Harden 58–62 HRC, grind Ø12 g6, Ra 0.4. Runs in PTFE-composite bushings: no grease			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

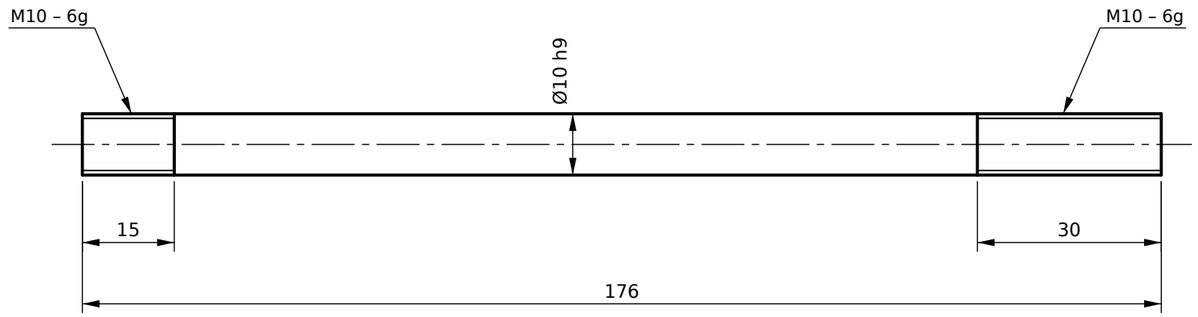


SIDE VIEW

NOTES

1. Retaining-ring grooves for DIN 471 – 16: groove Ø15.2 h12, width 1.1 H13, both ends.
2. Ends 0.5 × 45° chamfer. Groove inner edge at 36.1 from the pin centre (bearing faces at ±36).
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **B**	SHEET: 18 / 30	
TITLE: **PIN O (HANDLE PIVOT)**			
DRAWING NO.: **MP-117**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: 100Cr6 (GCr15) bar, or hardened ground dowel stock Ø16 g6			
HEAT TREATMENT / FINISH: Harden 58–62 HRC, grind Ø16 g6, Ra 0.4. Runs in PTFE-composite bushings: no grease			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

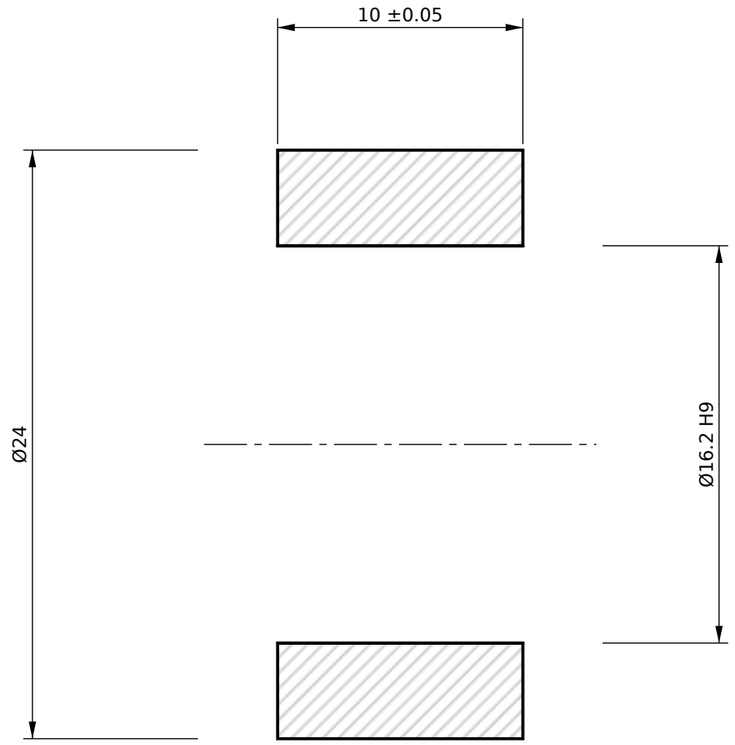


SIDE VIEW

NOTES

1. Bottom thread screws into the ram MP-109 (M10 × 15) with Loctite 243.
2. Rev B return spring: music wire Ø3.5 (EN 10270-1 DH), OD 31.5, free length 122, ≈ 16 active coils, ends closed and ground, shot-peened.
3. Rate ≈ 4.2 N/mm. Installed 109.5 at handle-up (≈ 60 N); 70.6 at handle-down (≈ 225 N, τ ≈ 445 MPa). Solid length ≤ 63.
4. Top: Ø32 × 3 washer and M10 nylock nut. The two springs together must lift the handle from the fully-down position.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 19 / 30
TITLE: **RETURN ROD**			
DRAWING NO.: **MP-118**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: C45 Ø12 bright bar			
HEAT TREATMENT / FINISH: Zinc plate 8 µm			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

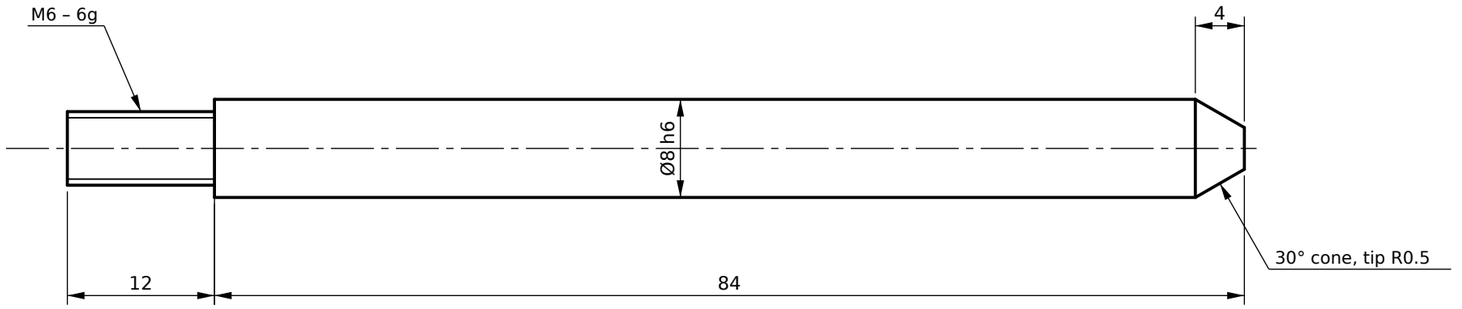


SECTION

NOTES

1. Rev B: bore Ø16.2 for the Ø16 pin O. Fits between the handle lever and each pivot cheek.
2. Faces parallel within 0.02.
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 20 / 30
TITLE: **PIVOT SPACER**			
DRAWING NO.: **MP-119**	QTY: 2	SCALE: 4:1	UNITS: mm
MATERIAL: CuSn8 bronze tube Ø26 × Ø14, or bar			
HEAT TREATMENT / FINISH: None			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED



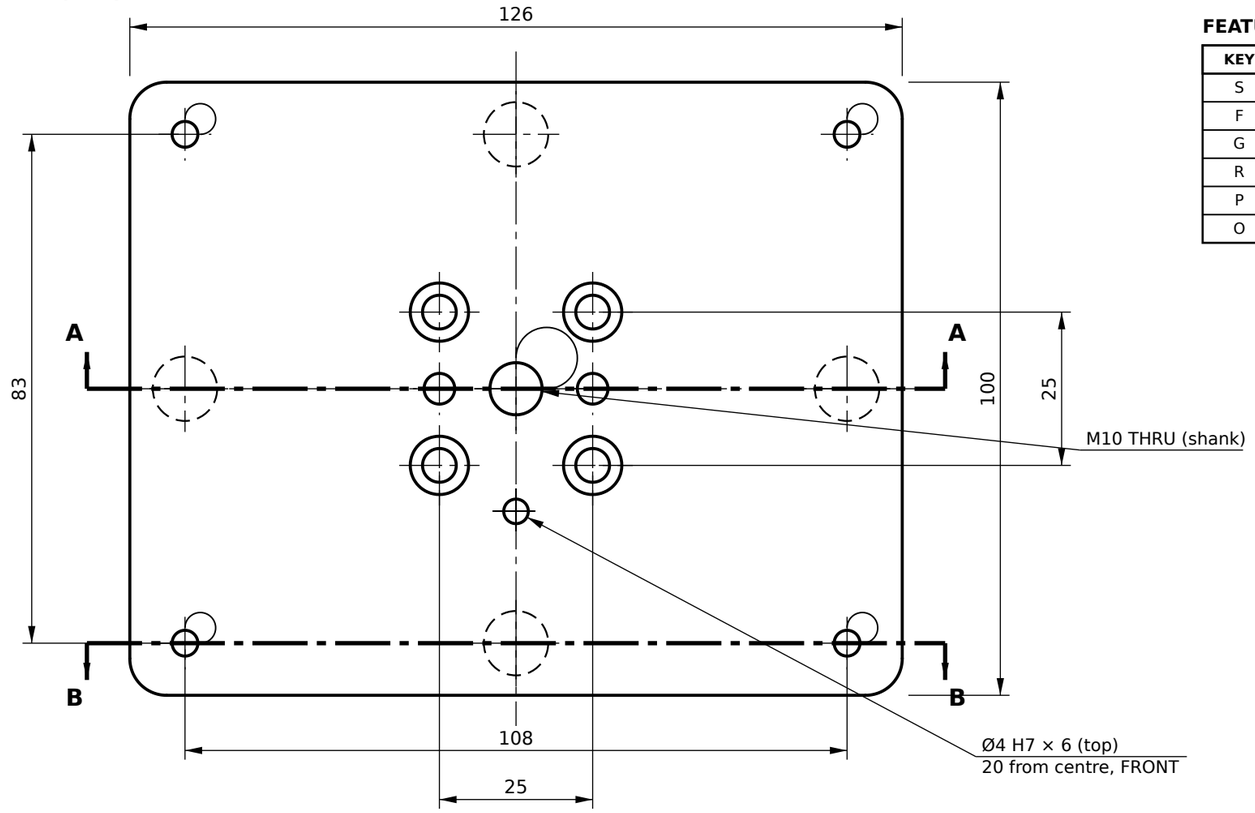
SIDE VIEW

NOTES

1. NEW in Rev B. Screwed into the ram underside (MP-109, hole PL) with Loctite 243; tip 84 below the ram.
2. Enters the die table hole PA or PB 6 mm before the dies touch. If the table is more than about 2 mm off a station, the tip lands on the table and stops the stroke with the dies still 6 mm apart.
3. Within ±2 mm, the 30° tip pulls the table onto the exact station (±0.02).
4. Tip: 30° half-angle cone, 4 long, tip radius R0.5, polished.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **B**	SHEET: 21 / 30	
TITLE: **STATION PILOT**			
DRAWING NO.: **MP-120**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: 100Cr6 (GCr15) Ø10 bar			
HEAT TREATMENT / FINISH: Harden 58–62 HRC, grind Ø8 h6, Ra 0.4			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom

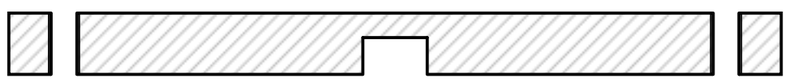


FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
S	centre	M10 THRU (shank MP-202)
F	(±12.5, ±12.5)	Ø5.5 THRU, CBORE Ø9.5 × 5.5 from the TOP (M5 × 20 ISO 4762)
G	(±12.5, 0)	Ø5 H7 THRU (dowels Ø5 m6 × 20)
R	(±54, ±41.5)	M5 THRU (frame shoulder screws ISO 7379 Ø6 × 10)
P	(±54, 0), (0, ±41.5)	Ø10.5 × 6 deep from the BOTTOM (spring pockets)
O	(0, +20)	Ø4 H7 × 6 deep from the TOP (orientation pin)



SECTION A–A

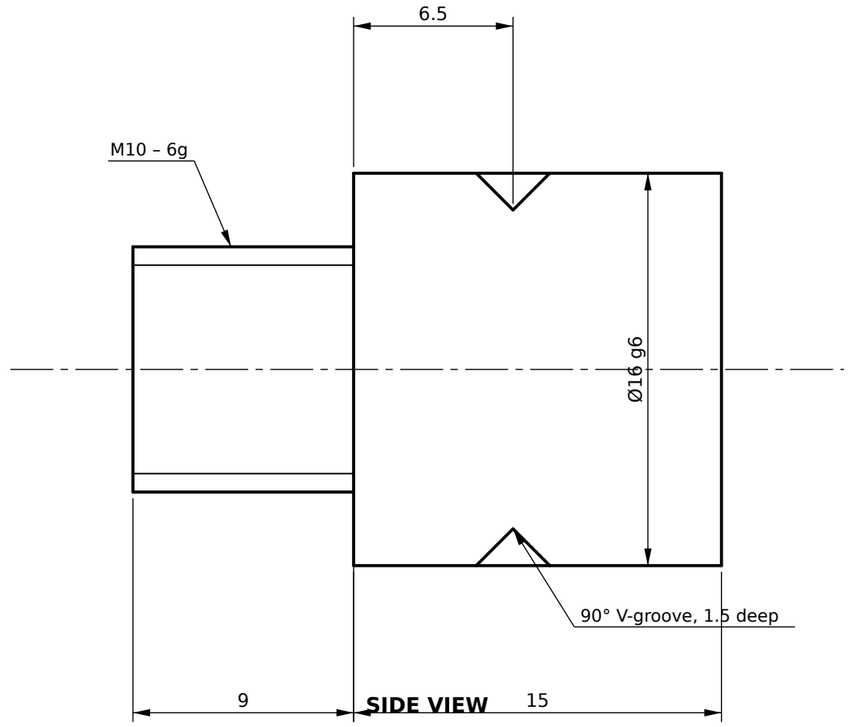


SECTION B–B (Z = ±41.5)

NOTES

1. Rev B universal part: identical for all four die sizes (5050, 6363, 5380, 6590). X = along the die table.
2. Mark each part on a side face with its drawing no. (and A/B where applicable).
3. Rev B: one body for every size (was X + 32 × Z + 32). Only the plug MP-203 and frame MP-204 change with size.
4. Top face (die seat against holder/shims) and bottom face parallel 0.01, flat 0.01.
5. Upper frame springs: 4 × ISO 10243 Ø10 × 25 blue. Cavity 22 at rest (12 % preload), 16.5 at full travel (34 %). Replace every 250 000 strokes.
6. Orientation pin hole Ø4 H7 × 6 deep in the top face, 20 from centre toward the FRONT; press in Ø4 m6 × 10.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 22 / 30
TITLE: **UPPER DIE BODY (UNIVERSAL)**			
DRAWING NO.: **MP-201**	QTY: 4 (1 per set)	SCALE: 1:1	UNITS: mm
MATERIAL: 1.2312 (P20+S) prehardened 28–32 HRC, or C45 QT			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED



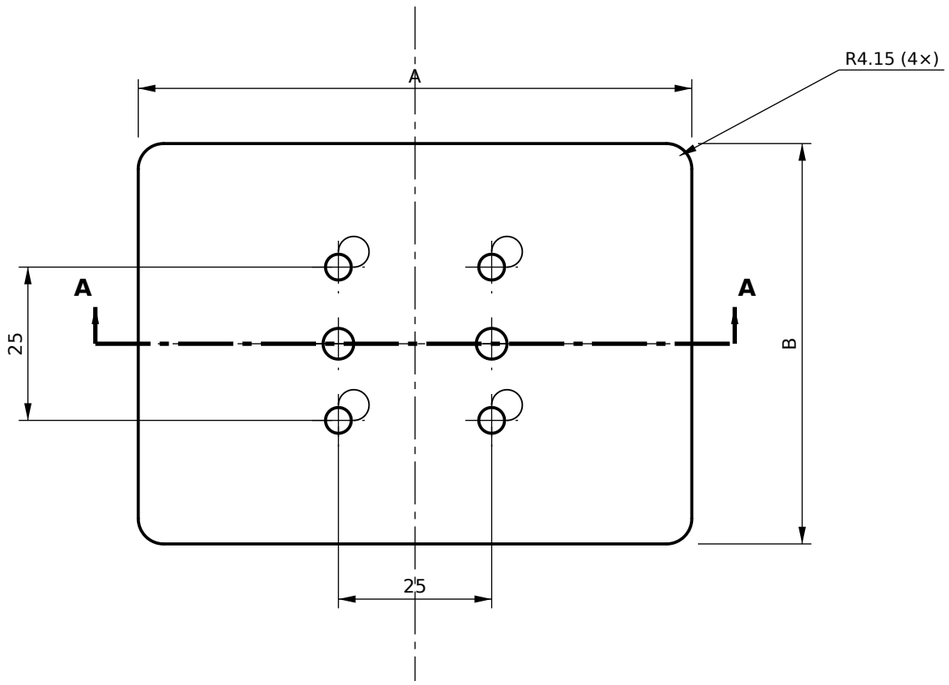
SIDE VIEW

NOTES

1. Screw into the die body MP-201 (M10 THRU) with Loctite 243 until the Ø16 face seats; face square to axis 0.01.
2. V-groove takes the cone-point lock screw in the die holder MP-111 (groove centre 6.5 from the seating face).
3. Top end chamfer 1 × 45° (lead-in).
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 23 / 30
TITLE: **UPPER DIE SHANK**			
DRAWING NO.: **MP-202**	QTY: 1 per die set (4)	SCALE: 4:1	UNITS: mm
MATERIAL: 42CrMo4 Ø20 bar, QT 28–32 HRC			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:

PLAN (TOP) — front at bottom



DIMENSION TABLE (mm)

SIZE	X	Z	A	B
5050	50	50	50.3	50.3
6363	63.5	63.5	63.8	63.8
5380	80	53	80.3	53.3
6590	90	65	90.3	65.3

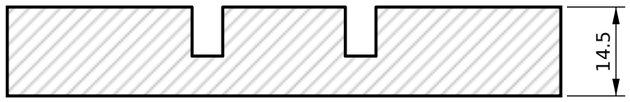
Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X + 0.30 (plug length, ±0.01)

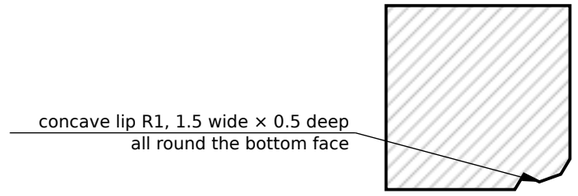
B = Z + 0.30 (plug width, ±0.01)

FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
F	(±12.5, ±12.5)	M5 × 10 deep from the TOP (4×)
G	(±12.5, 0)	Ø5 H7 × 8 deep from the TOP (dowels Ø5 m6)



SECTION A–A



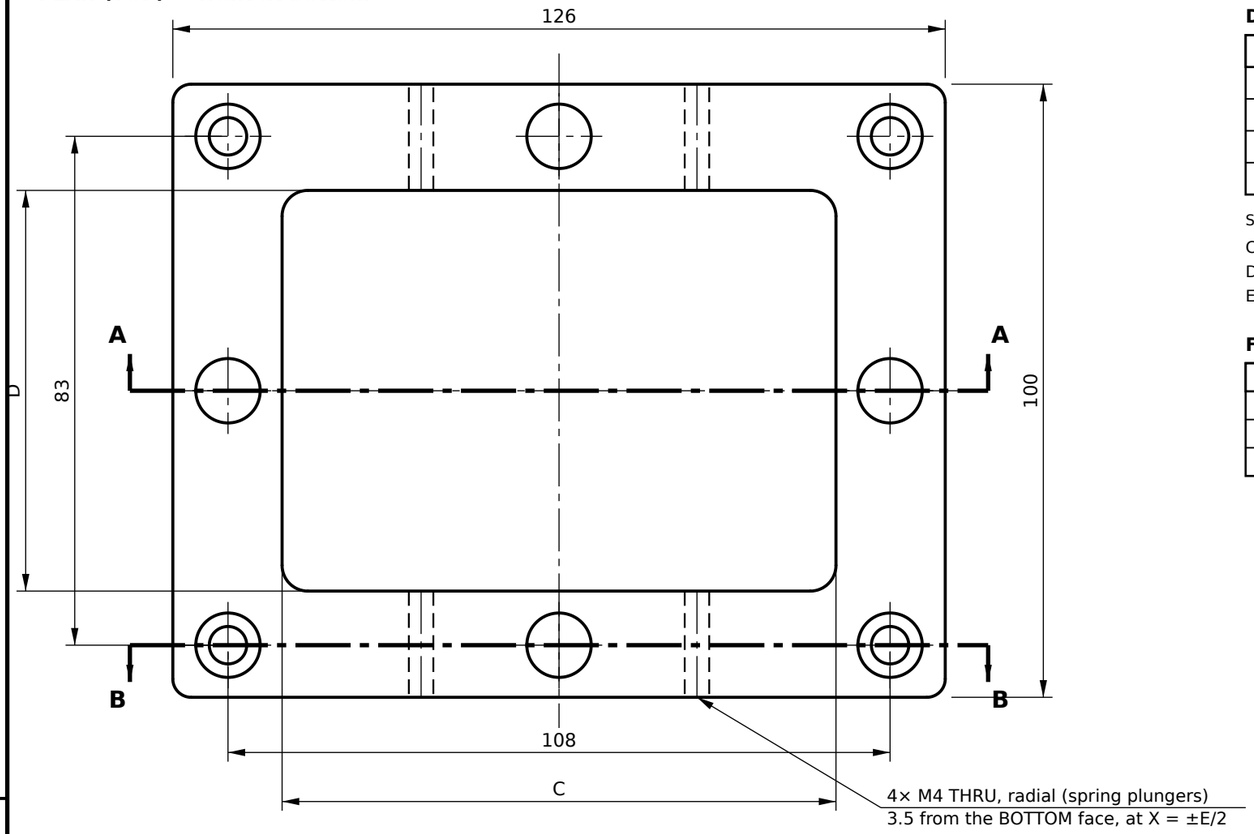
DETAIL Y (5:1) working edge

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Profile A × B with R4.15 corners: wire-EDM after hardening, ±0.01. Sliding fit in the upper frame opening (frame = plug + 0.05).
4. Bottom (working) face flat 0.01, Ra 0.2, with lip per Detail Y. Fixing pattern 25 × 25 is the same for every size (Rev B).
5. Height 14.5 ±0.01: sets the 5.5 mm recess between plug face and frame face.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

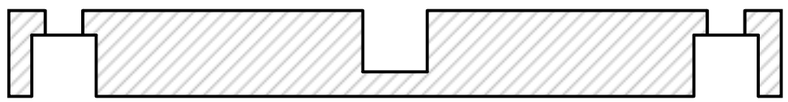
PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **B**	SHEET: 24 / 30	
TITLE: **UPPER DIE PLUG**			
DRAWING NO.: **MP-203**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype set; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom





SECTION A–A



SECTION B–B (Z = ±41.5)

DIMENSION TABLE (mm)

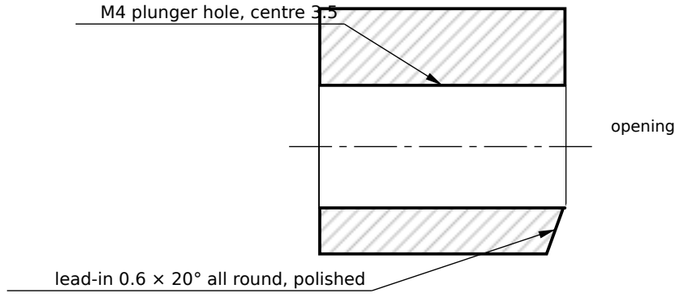
SIZE	X	Z	C	D	E
5050	50	50	50.35	50.35	25
6363	63.5	63.5	63.85	63.85	31.75
5380	80	53	80.35	53.35	40
6590	90	65	90.35	65.35	45

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

C = X + 0.35 (opening, +0.01/0)
D = Z + 0.35 (opening, +0.01/0)
E = X / 2 (pitch of the plungers in X)

FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
R	(±54, ±41.5)	Ø6.1 THRU, CBORE Ø10.5 × 10 from the BOTTOM
P	(±54, 0), (0, ±41.5)	Ø10.5 × 10 deep from the TOP (spring pockets)
N	(±E/2, ±(D/2 … 50))	M4 radial THRU from the long sides, 3.5 above the bottom face

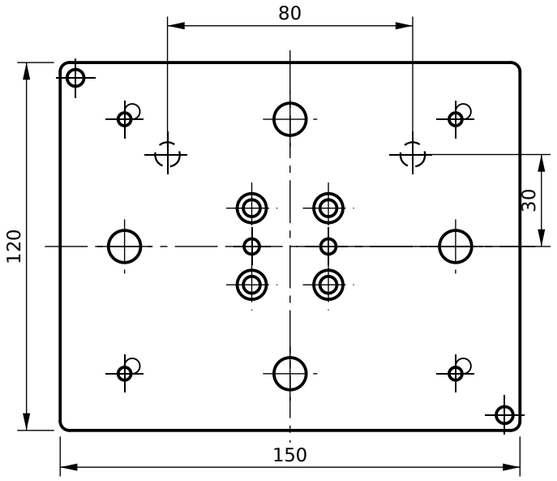


DETAIL P (5:1) bottom inner edge

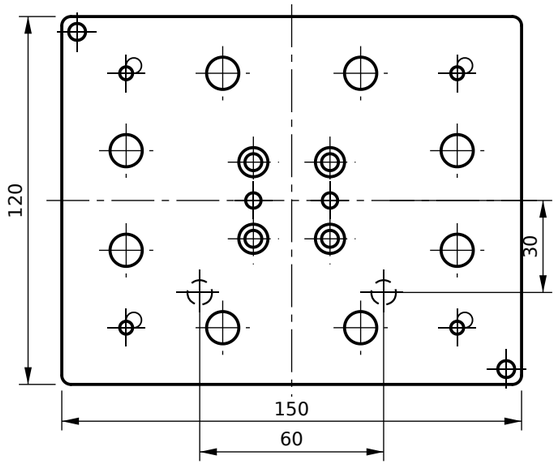
NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Rev B: outside 126 × 100 for every size. Opening C × D, R4.2, THRU: wire-EDM after hardening, lap Ra 0.2.
4. Rev B: bottom inner edge lead-in 0.6 × 20° (Detail P), and 4 nylon-tip spring plungers M4 (protrude 0.2–0.3 into the opening) so the wrapped shell cannot drop out.
5. Retention: 4 × Ø6.1 THRU, CBORE Ø10.5 × 10 from the BOTTOM. Spring pockets 4 × Ø10.5 × 10 from the TOP. Faces parallel 0.01.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **B**	SHEET: 25 / 30	
TITLE: **UPPER DIE FRAME**			
DRAWING NO.: **MP-204**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype set; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED



SHOE A — PLAN (TOP)



SHOE B — PLAN (TOP)

DA (shoe A): Ø8 H7 × 8 BOTTOM at (±40, −30) = back. DB (shoe B): at (±30, +30) = front.

FEATURES (from the shoe centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE	SHOE
R	(±54, ±41.5)	M5 THRU (frame shoulder screws)	A, B
F	(±12.5, ±12.5)	Ø5.5 THRU, CBORE Ø9.5 × 5.5 BOTTOM	A, B
G	(±12.5, 0)	Ø5 H7 THRU (pedestal dowels)	A, B
T	(70, 55), (−70, −55)	Ø5.5 THRU (thumb screws)	A, B
DA	(±40, −30)	Ø8 H7 × 8 BOTTOM (table dowels)	A
DB	(±30, +30)	Ø8 H7 × 8 BOTTOM (table dowels)	B
SA	(±54, 0), (0, ±41.5)	Ø10.5 × 6 TOP (4 spring pockets)	A
SB	(±54, ±16.25), (±22.5, ±41.5)	Ø10.5 × 6 TOP (8 spring pockets)	B

SECTION A–A, SHOE A, through Z = 0 (1:1)



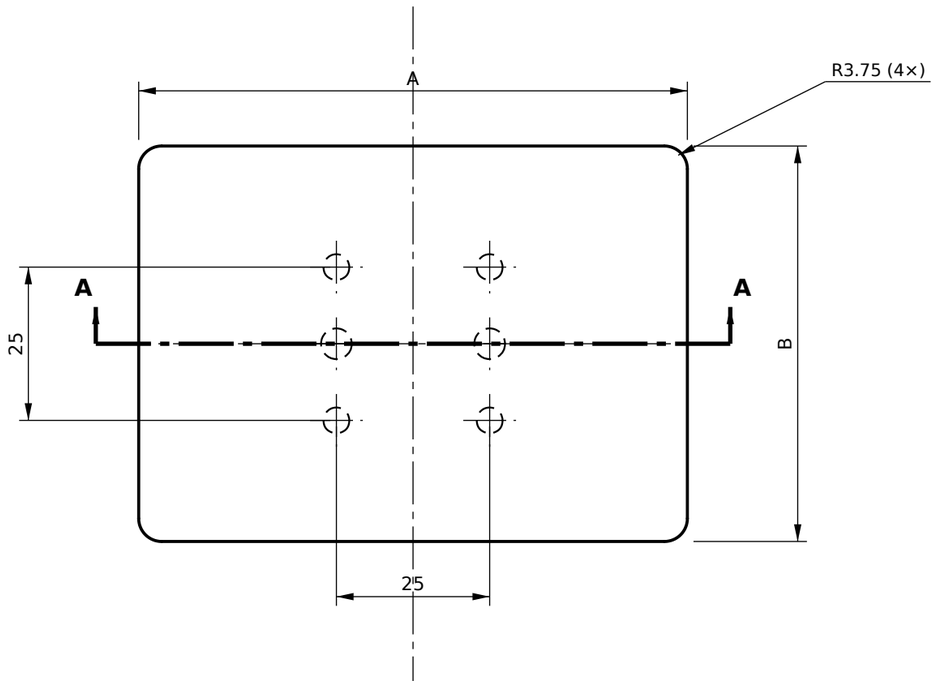
SECTION, SHOE B, through Z = +16.25 (1:1)

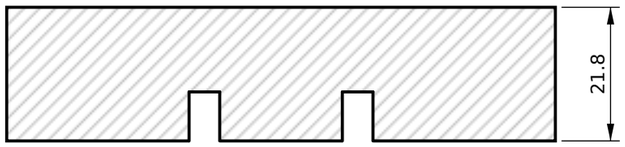
NOTES

1. Rev B universal part: identical for all four die sizes (5050, 6363, 5380, 6590). X = along the die table.
2. Mark each part on a side face with its drawing no. (and A/B where applicable).
3. Rev B: 150 × 120, the same for every size. A and B differ only in the table dowels (DA / DB) and the spring pockets (SA / SB).
4. Table dowels Ø8 m6 × 16 pressed into DA / DB from the BOTTOM, 8 protruding. A and B use different pitches so they cannot be swapped.
5. Thumb screws: M5 × 16 knurled, head Ø12 max (larger heads foul the frame corner or the stop block).
6. Lower springs: ISO 10243 Ø10 × 32 (A: 4 × green; B: 8 × red). Cavity 28 at rest (12.5 %), 20 at full travel (37.5 %). Replace every 250 000 strokes.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **B**	SHEET: 26 / 30
TITLE: **LOWER DIE SHOE A / B (UNIVERSAL)**			
DRAWING NO.: **MP-205**	QTY: A + B per set	SCALE: 1:2 / 1:1	UNITS: mm
MATERIAL: C45 plate 12 mm (blank 155 × 125), faced to 10; or 6061-T6			
HEAT TREATMENT / FINISH: Black oxide; paint front edge band: A green, B red			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:

PLAN (TOP) — front at bottom





SECTION A–A

DIMENSION TABLE (mm)

SIZE	X	Z	A	B
5050	50	50	49.5	49.5
6363	63.5	63.5	63	63
5380	80	53	79.5	52.5
6590	90	65	89.5	64.5

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X − 0.5 (±0.01)

B = Z − 0.5 (±0.01)

FEATURES (positions from the part centre, +Z = front)

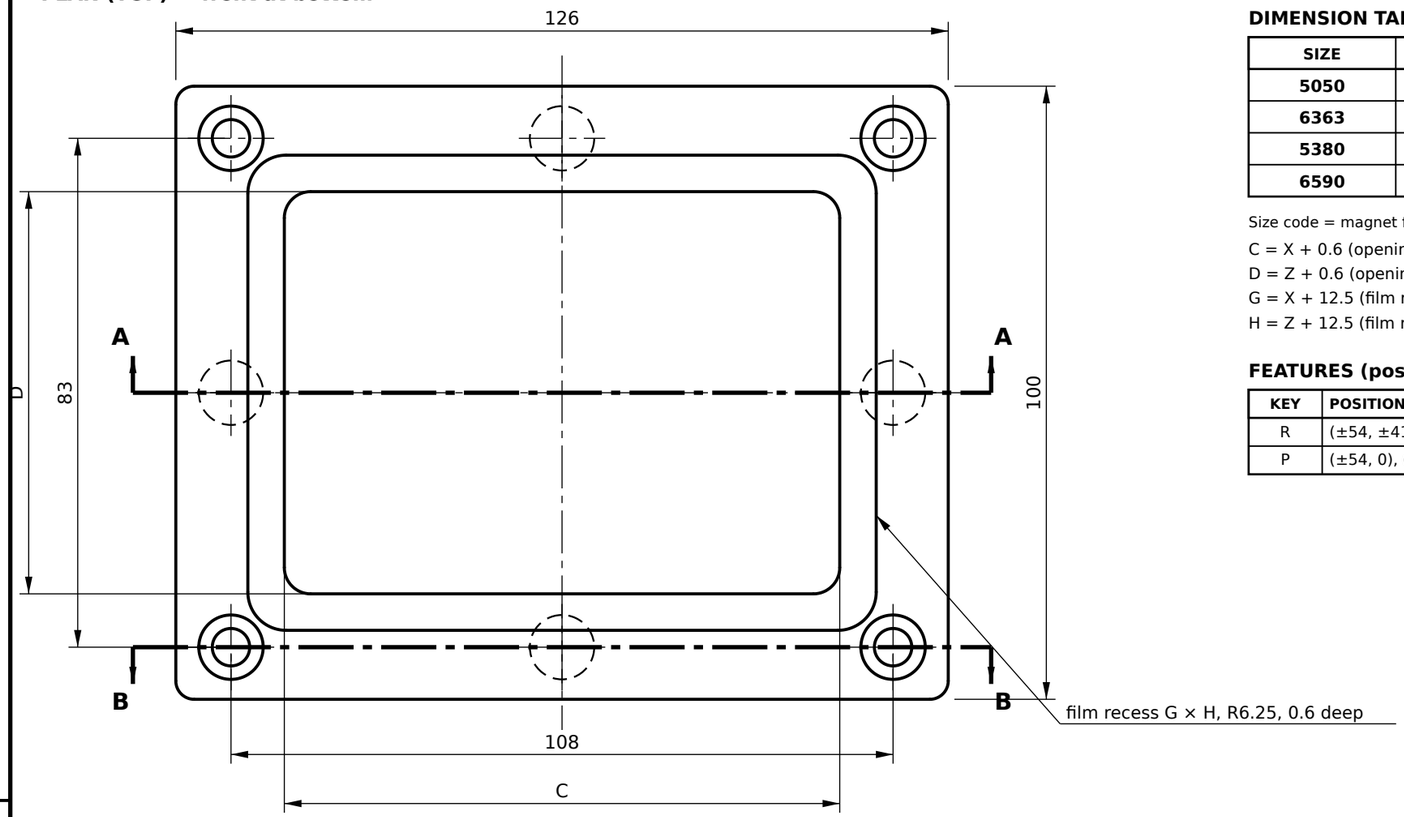
KEY	POSITIONS (X, Z)	FEATURE
F	(±12.5, ±12.5)	M5 × 10 deep from the BOTTOM (4×)
G	(±12.5, 0)	Ø5 H7 × 8 deep from the BOTTOM (dowels Ø5 m6)

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. The tinplate shell sits face-up over this pedestal. Top face flat 0.01, Ra 0.2; top edge R0.3.
4. Height 21.8 ±0.02. Fixing pattern 25 × 25 is the same for every size (Rev B).
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **B**	SHEET: 27 / 30	
TITLE: **PICKUP PEDESTAL (DIE A)**			
DRAWING NO.: **MP-206**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype set; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom



DIMENSION TABLE (mm)

SIZE	X	Z	C	D	G	H
5050	50	50	50.6	50.6	62.5	62.5
6363	63.5	63.5	64.1	64.1	76	76
5380	80	53	80.6	53.6	92.5	65.5
6590	90	65	90.6	65.6	102.5	77.5

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

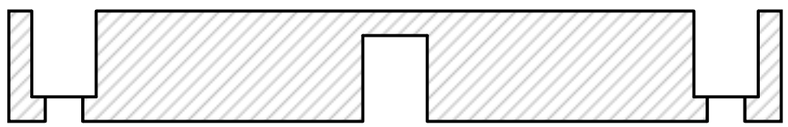
C = X + 0.6 (opening)
D = Z + 0.6 (opening)
G = X + 12.5 (film recess, 0.6 deep)
H = Z + 12.5 (film recess)

FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
R	(±54, ±41.5)	Ø6.1 THRU, CBORE Ø10.5 × 14 from the TOP
P	(±54, 0), (0, ±41.5)	Ø10.5 × 14 deep from the BOTTOM (spring pockets)



SECTION A–A



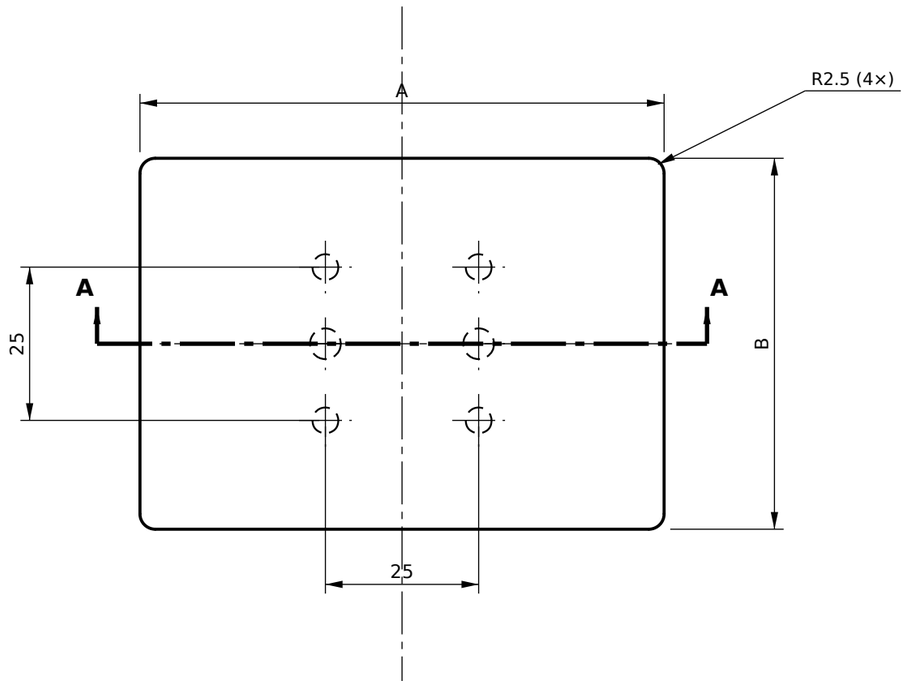
SECTION B–B (Z = ±41.5)

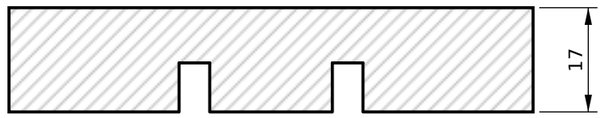
NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Rev B: outside 126 × 100 for every size. Opening C × D, R4.3, THRU, wire-EDM; top inner edge R0.5. Film recess G × H, R6.25, 0.6 deep, Ra 0.8.
4. Retention: 4 × Ø6.1 THRU, CBORE Ø10.5 × 14 from the TOP (ISO 7379 Ø6 × 12 into the shoe). Spring pockets 4 × Ø10.5 × 14 from the BOTTOM.
5. Frame travel 8: it bottoms on the shoe.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **B**	SHEET: 28 / 30	
TITLE: **PICKUP FRAME (DIE A)**			
DRAWING NO.: **MP-207**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype set; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom





SECTION A–A

DIMENSION TABLE (mm)

SIZE	X	Z	A	B
5050	50	50	45.5	45.5
6363	63.5	63.5	59	59
5380	80	53	75.5	48.5
6590	90	65	85.5	60.5

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X − 4.5 (±0.02)

B = Z − 4.5 (±0.02)

FEATURES (positions from the part centre, +Z = front)

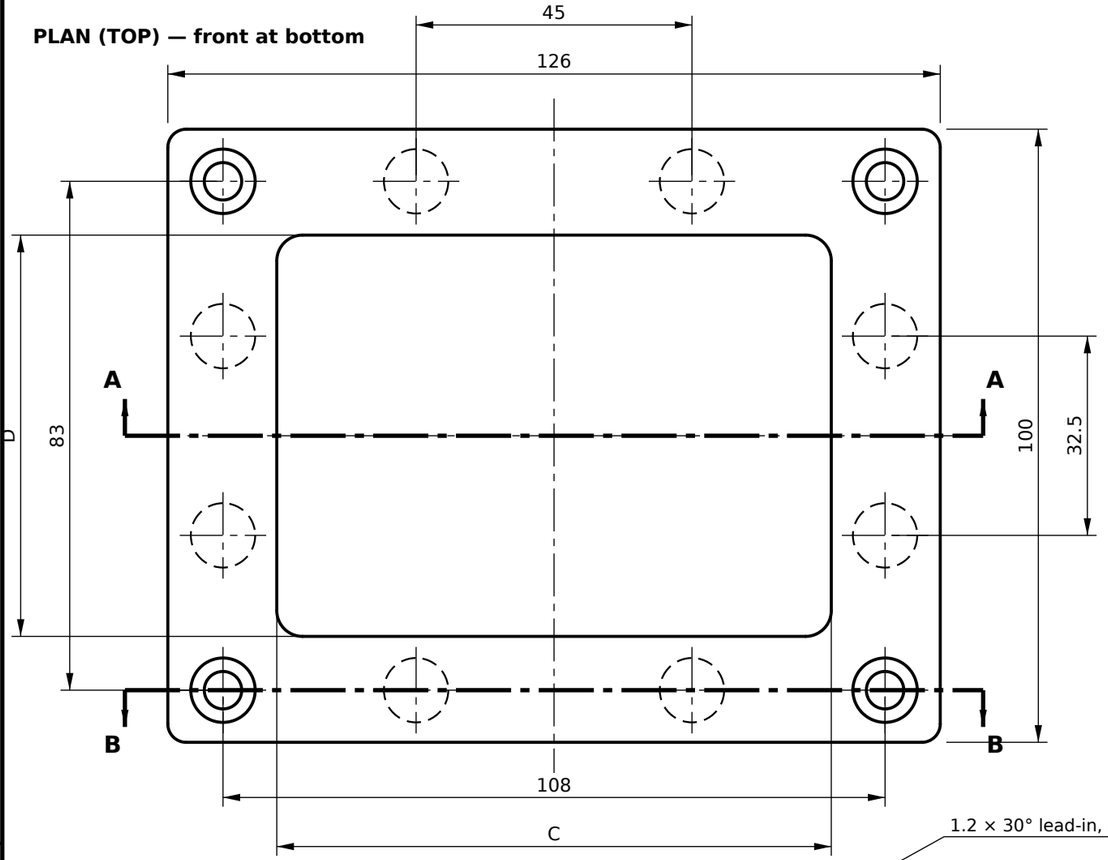
KEY	POSITIONS (X, Z)	FEATURE
F	(±12.5, ±12.5)	M5 × 10 deep from the BOTTOM (4×)
G	(±12.5, 0)	Ø5 H7 × 8 deep from the BOTTOM (dowels Ø5 m6)

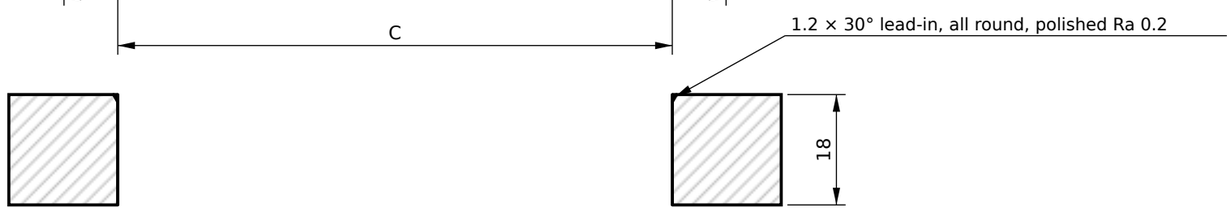
NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Smaller than the back plate (X − 1.4) so the shell edge can roll under the back. Top face flat 0.01, Ra 0.2; top edge R0.5.
4. Height 17.0 ±0.02. Fixing pattern 25 × 25 is the same for every size (Rev B).
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **B**	SHEET: 29 / 30	
TITLE: **CRIMP PEDESTAL (DIE B)**			
DRAWING NO.: **MP-208**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype set; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom





SECTION A–A



SECTION B–B (Z = ±41.5)

DIMENSION TABLE (mm)

SIZE	X	Z	C	D
5050	50	50	50.45	50.45
6363	63.5	63.5	63.95	63.95
5380	80	53	80.45	53.45
6590	90	65	90.45	65.45

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

C = X + 0.45 (opening, +0.01/0)

D = Z + 0.45 (opening, +0.01/0)

FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
R	(±54, ±41.5)	Ø6.1 THRU, CBORE Ø10.5 × 14 from the TOP
P	(±54, ±16.25), (±22.5, ±41.5)	Ø10.5 × 14 deep from the BOTTOM (8 spring pockets)

NOTES

1. Tabulated drawing: one part per die size. Shown to scale for size 6590 (65 × 90 magnet). X = long side, along the die table.
2. Mark each part on a side face: drawing no., size code and A/B where applicable (e.g. "MP-207 6590 A").
3. Rev B: outside 126 × 100 for every size. Opening C × D, R4.2, THRU, wire-EDM, lap Ra 0.2. Top lead-in 1.2 × 30°: folds the film under the shell edge.
4. Retention as MP-207. Spring pockets 8 × Ø10.5 × 14 from the BOTTOM. A drop of silicone release on the lead-in stops laser-printed film slipping.
5. Rev B: forming edge must have no EDM recast layer (2 skim cuts + re-temper): recast is where crimp frames crack.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **B**	SHEET: 30 / 30	
TITLE: **CRIMP FRAME (DIE B)**			
DRAWING NO.: **MP-209**	QTY: 1 per die set (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype set; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED